

# MPHBS: A novel hybrid metaheuristic with SARSA-guided information exchange

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## ABSTRACT

Hybrid metaheuristics can benefit from heterogeneous search processes, but their effectiveness depends on how information is exchanged between component optimizers. In this study, we propose Marine Predator Honey Badger SARSA (MPHBS), a hybrid Honey Badger Algorithm (HBA)–Marine Predators Algorithm (MPA) architecture in which a SARSA-style mediator regulates component-level information exchange between two independently evaluated populations. Rather than replacing the original HBA and MPA search operators, MPHBS ranks their candidate memories and organizes them into a tensorized decision structure indexed by problem dimension, candidate rank, and source population. SARSA acts as a lightweight on-policy mechanism that learns which source-ranked components should be reused during component-wise hybrid-vector assembly. To control evaluation cost, candidates are internally generated through proxy screening, but only one is evaluated with the expensive objective function per sub-exploration episode. MPHBS is assessed on CEC2022 and Finite Impulse Response (FIR) filter design problems against internal references, including MPHBS as the HBA–MPA hybrid without the mediator, and six representative external competitors. Internal validation selected B-C1 and A-C1 as final domain-specific configurations for CEC2022 and FIR, respectively. On CEC2022, B-C1 demonstrates robust competitiveness within the leading group while remaining statistically comparable to the MPHBS backbone. On FIR, A-C1 provides stronger empirical evidence with clearer advantages under the adopted filter-design protocol. These results suggest that the main strength of MPHBS lies in its information-exchange architecture, where the SARSA-style mediator provides a benefit that depends on the interaction between the search landscape and the recombination of ranked candidate memories.

## 1. Introduction

Metaheuristic optimization has become a fundamental tool for addressing black-box and highly constrained problems in transportation [1], logistics [2], energy [3], climate-related systems [4], social networks [5], healthcare [6], and Industry 4.0 [7]. These domains illustrate the broad need for robust optimizers when gradient information is unavailable, unreliable, or too costly to exploit directly [8]. The present study does not design domain-specific operators for all these application areas; instead, it focuses on a general continuous optimization mechanism and evaluates it on CEC2022 as a controlled benchmark suite and on Finite Impulse Response (FIR) filter design as a structured signal-processing problem. In such settings, practical performance depends not only on exploration capability, but also on avoiding premature convergence, escaping stagnation, and maintaining an effective exploration–exploitation balance under limited evaluation budgets [9].

To mitigate these limitations, ensemble, hybrid, and learning-based metaheuristics have received increasing attention [10–12]. Their main premise is that heterogeneous search behaviors can complement each other when useful information is exchanged in a controlled manner. Previous studies have considered several forms of information transfer, including population-level cooperation and information sharing [13,14], synthetic information models [10,15], perturbation-driven diversification [16], and clustering or mutation-based hybridization schemes [17]. These mechanisms are valuable, but they often operate at a coarse scale: complete operators are selected, elite individuals are injected, search parameters are tuned, or populations are merged globally. What remains comparatively underexplored is a learning-guided mechanism capable of exchanging information at the level of evaluated solution components, so that new candidates can be assembled dimension by dimension from complementary sources through

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component-wise hybrid-vector assembly, without replacing the original search dynamics of the base algorithms.

This gap is especially relevant when the search process begins to stagnate. In such cases, simply injecting randomness or restarting the population may recover diversity, but often at the cost of losing valuable structural information accumulated during the search. A more effective alternative is to sub-explore the updated solution matrices generated by different optimizers and selectively recombine their informative components. This enables the generation of new candidate solutions that are neither purely inherited from a single metaheuristic nor obtained through blind perturbation, but instead arise from an adaptive and informed interaction between multiple search processes.

Motivated by this perspective, this study considers two population-based optimizers with complementary search behaviors: HBA [18] and MPA [19]. HBA alternates between digging and honey-search behaviors around promising regions, whereas MPA uses predator-prey-inspired movement phases to balance exploration and exploitation. Both algorithms are population-based and best-guided, which makes their evaluated populations suitable as candidate memories for a shared information-exchange mechanism. Their use in electrical engineering, deep learning, and feature selection also indicates that they are sufficiently general to serve as base optimizers in a continuous black-box setting [20–22]. Rather than treating HBA and MPA as isolated optimizers or replacing their native operators, we exploit their updated populations as two complementary information sources and introduce a mediator that learns how to recombine their ranked components during the search.

In this study, a hybrid optimization method called MPHBS is proposed, which integrates MPA, HBA, and a SARSA-style reinforcement-learning mediator. SARSA is adopted because it provides a lightweight on-policy learning rule that can update preferences from the actions actually selected during the search, without requiring a deep model or an additional training phase. In MPHBS, the reinforcement-learning component is not used to choose a complete optimizer or replace an operator. Instead, states and actions are associated with source-ranked component choices over active dimensions, and the reward is computed from the fitness change obtained after evaluating the assembled hybrid candidate. Thus, the learning process directly regulates information exchange between HBA and MPA. The central premise is that useful search information is not necessarily preserved only in the current global best solution, but may also remain distributed as historically informative components across the evolving populations of the base optimizers.

In this way, MPHBS transforms population history into actionable search guidance. The distinctive focus of the proposed architecture is the scale at which information is represented and exchanged: rather than traversing the candidate memory blindly or relying on coarse operator-level hybridization, the SARSA-style mediator learns which source–rank combinations are worth exploiting for each dimension. This enables a selective and informed information-exchange process over ranked HBA and MPA candidate memories. The approach preserves the intrinsic search logic of the underlying metaheuristics while increasing the amount of useful exploration performed per expensive function evaluation.

The proposed architecture is evaluated through a two-stage experimental design. First, internal sensitivity and ablation analyses identify domain-specific final configurations of the same MPHBS mechanism: B-C1 for CEC2022 and A-C1 for FIR. These labels do not denote different algorithms; they denote validated settings of the same SARSA-guided information-exchange architecture under different benchmark domains. Second, the final configurations are compared against internal references and external competitors selected to represent different search families, including hybrid DE-based, learning-based, swarm-based, population-based, and physics-inspired mechanisms.

The main contributions of this article are as follows:

- We propose MPHBS, a learning-guided ensemble metaheuristic that coordinates the interaction between MPA and HBA through a SARSA-style mediator. Unlike conventional hybrid schemes that exchange information at the population or operator level, MPHBS performs selective component-wise recombination, aiming to improve search diversification, reduce premature stagnation, and preserve the complementary search dynamics of the base optimizers.
- We introduce a tensorized decision representation for the reinforcement-learning component, organized across problem dimensions, ranked candidate indices, and source populations. This formulation enables fine-grained, dimension-wise action selection and avoids blind traversal of the candidate memory, allowing the mediator to learn which source–rank combinations are worth exploiting for each dimension.
- We develop a proxy-screened sub-exploration mechanism in which  $K$  low-cost hybrid candidates are internally generated through component-wise hybrid-vector assembly at each episode, but only the proxy-best candidate is evaluated with the expensive objective function. This one-expensive-evaluation-per-episode design increases the amount of candidate exploration per function evaluation and constitutes a budget-aware alternative to exhaustive or naive action scanning.
- We introduce a dimension-wise guided mutation mechanism in which solution components are selectively replaced using ranked candidate memories extracted from the independently evolved HBA and MPA populations. Instead of applying multiple external operators to the entire population, the proposed mechanism performs targeted component substitution over active dimensions, enabling historically informative solution fragments from the base optimizers to be selectively recovered and reused whenever they remain beneficial for the ongoing search.
- We design a conservative SARSA-style update rule that propagates learning only through the actions involved in the proxy-winning candidate and only over the currently active dimensions. Accepted transitions bootstrap from the selected successor actions, whereas rejected transitions retain the current action values in the bootstrap term. This keeps learning tied to the actually evaluated candidate while preventing unproductive state replacements.
- We conduct a comprehensive empirical validation on the CEC2022 benchmark suite and FIR filter design problems, including sensitivity analysis, ablation studies, convergence analysis, and comparisons against internal baselines, parent optimizers, and six external competitors: Hybrid Equilibrium-Differential Evolution (EODE) [23,24], Reinforcement-Learning Teaching-Learning-Based Optimization (RLTLBO) [25], Quasi-opposition and Q-learning Marine Predators Algorithm (QLMPA) [26], Grey Wolf Optimizer (GWO) [27], fitness-distance-balance teaching-learning artificial bee colony (FDB-TLABC) [28], and Fourier Transform Optimizer (FTO) [29].

The remainder of this article is organized as follows. Section 2 reviews related work and positions the proposed method with respect to existing hybrid and ensemble metaheuristics. Section 3 introduces the preliminaries required to understand the proposed framework. Section 4 presents the development of MPHBS in detail. Section 5 reports the experimental protocol and results. Section 6 discusses the main findings, Section 7 analyzes threats to validity, Section 8 summarizes reproducibility aspects, and Section 9 concludes the paper and outlines future research directions.

## 2. Related work

Before reviewing specific studies, it is useful to clarify how the terms hybrid and ensemble are used in this manuscript. Hybrid metaheuristics are understood as optimization frameworks that combine

complementary algorithmic components, such as multiple metaheuristics, local search procedures, exact methods, learning mechanisms, or problem-specific operators, within a coordinated search process [30–32]. Ensemble metaheuristics, in contrast, emphasize the coordinated use of multiple algorithms, populations, agents, or operator pools as a collective search system, aiming to exploit complementary search behaviors to improve robustness, convergence speed, and solution quality [10,13]. This distinction is not absolute, because several recent learning-based methods combine both views. In this review, methods are discussed as hybrid when the emphasis is on tight integration within a single search architecture, and as ensemble when the emphasis is on cooperation among multiple search components. This distinction is important for positioning MPHBS: the proposed method is a hybrid HBA–MPA architecture, but its central contribution is an information-exchange mediator that operates over ranked candidate memories rather than over complete operators or complete algorithms.

### 2.1. Hybrid metaheuristic algorithms

Following this terminology, recent hybrid metaheuristics have increasingly combined heterogeneous search behaviors with adaptive or learning-based mechanisms to enhance exploration, exploitation, or information exchange among agents [11–14]. These approaches support robust search by coordinating information sharing, operator behavior, or population-level cooperation across complex optimization and application-oriented tasks [1–3,6]. Recent surveys, such as Yang et al. [33], also show that Q-learning and reinforcement-learning mechanisms are increasingly used to adapt metaheuristic search behavior. However, the level at which learning acts is important. In most reinforcement-learning-assisted hybrids, the learning component selects operators, adapts parameters, chooses local search procedures, or controls algorithmic modes, rather than regulating how evaluated solution components are reused across dimensions.

Regarding optimization algorithms involving learning-based methods, Samma et al. [34] introduced a Reinforcement Learning-based Memetic Particle Swarm Optimization model in which reinforcement learning controls particle operations such as exploration, convergence, high-jump, low-jump, and fine-tuning.

Seyyedabbasi et al. [11] used Q-learning to control exploration and exploitation in metaheuristic algorithms such as Incremental Grey Wolf Optimizer, Expanded Grey Wolf Optimizer, and Whale Optimization Algorithm.

Similarly, Niu et al. [35] proposed a Q-learning-based bacterial foraging optimization algorithm to guide step-size behavior. These works show that learning can improve the adaptive behavior of a metaheuristic, but the learned decisions are mainly associated with operator choice, search mode, or control parameters.

Other studies have used learning mechanisms to coordinate complete algorithms, local search operators, or parameter settings. Wang et al. [36] combined Genetic Algorithm, Artificial Bee Colony, Shuffled Frog-Leaping Algorithm, and Simulated Annealing through a Q-learning-based selection mechanism for integrated disassembly and preprocessing scheduling.

Cui et al. [37] proposed a reinforcement-learning-based hybrid Genetic Algorithm in which Expected SARSA guides parameter selection during population evolution.

Ma et al. [38] integrated Q-learning with local search operators for unmanned surface vehicle scheduling, while Liu et al. [39] combined reinforcement learning and Particle Swarm Optimization for path planning. Huang et al. [40] incorporated reinforcement-learning strategies, adaptive weights, and dimensional learning into Particle Swarm Optimization, and Klein et al. [41] used reinforcement learning to optimize swarm-algorithm parameters. These studies are relevant because they confirm the usefulness of reinforcement learning in hybrid metaheuristics, but they still treat the learned decision as a supervisory choice over a predefined search action.

The distinction with MPHBS is therefore not the use of reinforcement learning alone, but the scale at which the learning decision is applied. In MPHBS, SARSA does not choose which base optimizer should run, nor does it replace the HBA or MPA update equations. Instead, the learning process is connected to the information-exchange mechanism: states and actions encode source-ranked component choices over active dimensions, and the reward is computed from the fitness change obtained after evaluating the assembled hybrid candidate. Thus, SARSA mediates which ranked HBA/MPA components are reused when constructing a candidate vector, rather than selecting a complete optimizer, a global operator, or a parameter setting.

Recent works on information exchange and component-level learning further show that cooperation among search agents remains an active research direction. Jia et al. [42] proposed an Experience Exchange Strategy in which individuals share experience to improve metaheuristic behavior. Huang et al. [43] introduced a crisscross-strategy-boosted optimizer that promotes information exchange across solution dimensions. Chen et al. [44] proposed an enhanced escape algorithm based on dimension-wise comprehensive learning and Cauchy–Gaussian mutation, showing the relevance of dimension-specific guidance for maintaining population diversity. Mao et al. [45] incorporated elite information sharing within a multi-strategy enhanced optimizer. These studies are close to the motivation of MPHBS because they recognize that search information may be distributed across individuals or dimensions. However, these approaches still differ from MPHBS in the level at which information is selected and reused. In MPHBS, information exchange is not implemented as a global migration rule, an operator switch, or a population-level communication mechanism; instead, evaluated HBA and MPA candidates are ranked, organized by source and dimension, and reused through SARSA-style component selection during hybrid-vector assembly.

Several recent IPA and UAV path-planning studies provide useful comparison points because they modify how search information is generated, diversified, or transferred in engineering-oriented settings. The Levy-flight-assisted IPA of Aslan et al. [46] updates the plasma extraction mechanism of IPA through Levy flight, mainly improving stochastic diversification within the IPA dynamics. This is a useful exploration-oriented modification, but it acts as a movement-diversification mechanism rather than as a learned information-exchange layer. By contrast, MPHBS keeps the original HBA and MPA update equations unchanged and introduces learning only at the stage where evaluated candidate information is ranked and recombined.

The GINT-IPA method of Aslan and Erkin [47] follows a different route by combining a greedy initializer with a redesigned IPA treatment schema. Its strength comes from starting the search with qualified UAV path candidates and from adapting the IPA treatment process to the geometry and objectives of the path-planning problem. The multithreaded Back-and-Forth algorithm [48] is even more explicitly problem-informed: it exploits the geometric structure of UAV path planning through parallel threads with different search bounds and can obtain competitive paths with fewer objective-function calls. These methods show the value of embedding domain knowledge into the search process. MPHBS addresses a complementary setting: it does not assume path geometry, qualified route initializers, or UAV-specific constraints, but instead learns how to reuse ranked components produced by two continuous optimizers.

Other IPA variants are especially relevant for distinguishing MPHBS from population-level information-transfer strategies. The multi-population IPA of Aslan and Erkin [49] extends IPA with multiple populations, migration-based information transfer, and modified donor usage. DUALIPA [50] executes two IPA populations simultaneously, initializes them from two greedy path candidates, and exchanges information through emigrant communication. These designs demonstrate that multi-population communication can improve path-planning performance, but the exchanged information remains mainly at the population, migration, or path-candidate level. MPHBS lowers this

interaction scale: the mediator selects source-aware ranked components for active dimensions and assembles a hybrid candidate before a single expensive objective evaluation. The IPA-based UCAV study [51] further illustrates the relevance of metaheuristics in real engineering path-planning problems; in this manuscript, the FIR benchmark plays a complementary role as a structured signal-processing domain in which partial candidate components can retain meaningful design information.

Beyond learning-based approaches, several other hybridization strategies have also been explored in the literature. Neto et al. [52] proposed a multi-agent metaheuristic hybridization framework for the automatic design of ensemble systems, whereas Li et al. [53] developed a hybrid metaheuristic for task assignment in parallel and distributed systems. Sharma et al. [54] combined Augmented Grey Wolf Optimizer and Cuckoo Search for global optimization, and Pereira et al. [55] introduced the Lichtenberg Algorithm as a hybrid physics-based metaheuristic. Wang et al. [56] combined butterfly and flower pollination mechanisms through mutualism, Mahajan et al. [57] fused arithmetic and grasshopper optimization mechanisms, and Saadi et al. [58] integrated improved manta ray foraging optimization with tabu search for UAV placement. These studies demonstrate that hybridization can improve search behavior by combining complementary operators, physical analogies, local search mechanisms, or problem-specific search components. However, their integration is mainly performed at the algorithmic, operator, or population level. MPHBS addresses a different interaction scale by regulating how evaluated components from two independently evolved populations are ranked, selected, and recombined dimension by dimension while preserving the original HBA and MPA search dynamics.

## 2.2. Ensemble metaheuristic algorithms

Within the ensemble view, learning-based mechanisms have increasingly been incorporated to enable dynamic cooperation, adaptive parameter control, and intelligent information sharing among metaheuristics [11,12,14,33]. These learning-driven ensembles often integrate reinforcement learning, supervised learning, or adaptive strategies to guide the interaction between component optimizers, resulting in enhanced exploration–exploitation balance across complex optimization tasks. A common pattern is that learning selects among local search operators, complete algorithms, or problem-specific improvement rules. This is effective for coordinating heterogeneous search components, but it differs from learning how to recombine evaluated solution fragments inside a continuous candidate vector.

For instance, Liu et al. [59] proposed an ensemble algorithm using Iterated Greedy, Artificial Bee Colony, Genetic Algorithm, and Differential Evolution, with Q-learning used to select among local search strategies.

Potharlanka et al. [60] integrated Particle Swarm Optimization, Firefly Algorithm, and Whale Optimization Algorithm with Deep Q-Learning for ensemble-based feature selection. Lin et al. [61] proposed an ensemble metaheuristic framework with Q-learning-based local search for urban traffic-light scheduling, while Gao et al. [62] used Q-learning to select local operators in a multi-algorithm framework for unmanned surface vehicle scheduling. Wang et al. [63] also integrated Q-learning with multiple population-based optimizers for urban traffic-light scheduling, and Yu et al. [64] employed several metaheuristics and local search strategies for surgery scheduling and rescheduling. These examples illustrate a strong application-driven trend: learning improves the coordination of several search components, but the learned policy usually acts over monolithic local search rules, complete optimizers, or domain-specific improvement operators.

From a critical perspective, this macro-level coordination can miss useful partial information already contained in individual candidate solutions. If one source population contains informative components for some dimensions and another source population contains better

components for different dimensions, selecting only a full optimizer or a full local search operator may discard part of that information. MPHBS targets this finer interaction scale. Rather than treating HBA and MPA as interchangeable complete optimizers, it keeps both search processes active and uses their ranked memories as component sources. SARSA then guides which source-ranked components are selected across active dimensions, allowing the algorithm to recombine partial information while preserving the original population dynamics of both base optimizers.

In addition to learning-based ensembles, several conventional ensemble metaheuristics have also been proposed in the literature. Yu et al. [65] introduced an Ensemble Grey Wolf Optimizer for image segmentation, while Jia et al. [66] proposed an ensemble mutation strategy for Remora Optimization. Wu et al. [67] combined metaheuristic and exact search within a divide-and-conquer framework for multisatellite observation scheduling, and Perez-Aracil et al. [68] used a multi-method ensemble metaheuristic for submerged arch design. Xue et al. [69] proposed an ensemble algorithm with self-adaptive learning population search techniques for high-dimensional optimization, whereas Xue et al. [70] combined Differential Evolution and Adam for neural network training. Perez-Aracil et al. [71] later introduced dynamic multi-method ensembles based on CRO-SL, and Palomo-Alonso et al. [72] presented TensorCRO as a TensorFlow-based implementation of a multi-method ensemble.

These methods show the value of cooperation among search processes, but they usually coordinate complete algorithms, operators, populations, substrate layers, or optimizer-specific update rules. The gap addressed here is finer-grained: MPHBS learns how to reuse ranked candidate components dimension by dimension while preserving the original dynamics of HBA and MPA.

## 2.3. External comparator algorithms

The external comparison uses representative algorithms that cover distinct search mechanisms rather than variants of the proposed architecture. GWO is included as a classical swarm-based optimizer with a widely recognized leadership hierarchy and encircling mechanism [27]. EODE is included as a recent hybrid differential evolution (DE)-based comparator, providing a non-trivial reference from the DE-related family of search strategies [23,24]. RLTLBO [25] and QQLMPA [26] are included as additional learning-based metaheuristic comparators, allowing the analysis to distinguish the proposed SARSA-guided component-exchange mechanism from the broader idea of adding learning to a population-based optimizer. FDB-TLABC is included as a recent population-based alternative with a search mechanism distinct from both GWO and DE-based hybrids [28]. Finally, FTO is included as a recent physics-inspired optimizer based on Fourier-transform search principles [29]. This comparator set is not intended to exhaustively rank all available metaheuristics; rather, it provides classical, hybrid DE-based, learning-based, recent population-based, and physics-inspired references for evaluating whether the final MPHBS configurations remain competitive outside the immediate HBA–MPA design space.

Table 1 summarizes the methodological position of MPHBS with respect to representative parent, hybrid, ensemble, learning-guided, and external comparator algorithms. The table is not intended as a performance ranking; instead, it highlights the search level at which each method operates and clarifies that the distinctive feature of MPHBS is the SARSA-guided reuse of ranked HBA/MPA components at the dimension level.

## 3. Preliminaries

### 3.1. HBA

HBA is inspired by the foraging behavior of the honey badger, a solitary mammal recognized for its aggressive and adaptive hunting

**Table 1**

Feature-based comparison between representative related metaheuristics and MPHBS. Checkmarks indicate that the corresponding methodological feature is explicitly present in the cited work or method group; blank cells indicate that the feature is not a defining component of that group. The comparison focuses on architecture rather than numerical performance.

| Method or group                                    | Involved references                             | Hybrid or ensemble coordination | RL or adaptive guidance | Operator/parameter-level control | Population-level information exchange | Dimension-wise guidance/reuse | Component-wise hybrid-vector assembly | Ranked source-aware memory | Proxy-screened single-FE episode |
|----------------------------------------------------|-------------------------------------------------|---------------------------------|-------------------------|----------------------------------|---------------------------------------|-------------------------------|---------------------------------------|----------------------------|----------------------------------|
| Single algorithms                                  | HBA [18],<br>MPA [19],<br>GWO [27],<br>FTO [29] |                                 |                         |                                  |                                       |                               |                                       |                            |                                  |
| Internal hybrid backbone                           | MPHB, based on HBA [18] and MPA [19]            | ✓                               |                         |                                  |                                       |                               |                                       |                            |                                  |
| External hybrid comparators                        | EODE [23,24],<br>FDB-TLABC [28]                 | ✓                               |                         |                                  |                                       |                               |                                       |                            |                                  |
| RL-based external comparators                      | RLTLBO [25],<br>QQLMPA [26]                     |                                 | ✓                       | ✓                                |                                       |                               |                                       |                            |                                  |
| RL-based hybrid metaheuristics                     | [11,36–38,40,41]                                | ✓                               | ✓                       | ✓                                |                                       |                               |                                       |                            |                                  |
| Learning/adaptive hybrid methods                   | [73]                                            | ✓                               | ✓                       |                                  |                                       |                               |                                       |                            |                                  |
| RL-based ensemble metaheuristics                   | [59,60,62–64]                                   | ✓                               | ✓                       | ✓                                |                                       |                               |                                       |                            |                                  |
| Information feedback/exchange methods              | [13–17,42]                                      |                                 | ✓                       |                                  | ✓                                     |                               |                                       |                            |                                  |
| Recent dimension-guided methods                    | [43–45]                                         |                                 | ✓                       | ✓                                |                                       | ✓                             |                                       |                            |                                  |
| Problem-informed IPA and UAV path-planning methods | [46–51]                                         | ✓                               |                         | ✓                                | ✓                                     |                               |                                       |                            |                                  |
| Proposed SARSA-guided hybrid                       | MPHBS                                           | ✓                               | ✓                       |                                  | ✓                                     | ✓                             | ✓                                     | ✓                          | ✓                                |

strategies [18]. In optimization terms, HBA alternates between two complementary search modes: a digging mode, which promotes exploratory movements guided by prey intensity, and a honey mode, which produces a more direct attraction toward the best solution found so far. In the version used within MPHBS, HBA evolves a population of candidate solutions and updates each agent with respect to the current best position of the HBA population.

The HBA population is represented by the candidate matrix in Eq. (1), where  $N$  denotes the population size and  $Dim$  is the dimensionality of the optimization problem:

$$X_{HBA} = \begin{bmatrix} X_{HBA,11} & X_{HBA,12} & \cdots & X_{HBA,1Dim} \\ X_{HBA,21} & X_{HBA,22} & \cdots & X_{HBA,2Dim} \\ \vdots & \vdots & \ddots & \vdots \\ X_{HBA,N1} & X_{HBA,N2} & \cdots & X_{HBA,NDim} \end{bmatrix} \quad (1)$$

Each row of  $X_{HBA}$  represents one  $Dim$ -dimensional candidate.

The initial candidate positions are generated uniformly within the search bounds:

$$X_{HBA,ij} = lb_j + r_{ij}(ub_j - lb_j), \quad (2)$$

where  $r_{ij} \sim U(0, 1)$ , and  $lb_j$  and  $ub_j$  denote the lower and upper bounds of the  $j$ th decision variable.

At each iteration, HBA stores the best solution found so far, denoted by  $X_{HBA,best}$ . The search is then guided by a light-intensity mechanism. For the  $i$ th agent, the intensity is computed as

$$I_i = r_2 \frac{S_i}{4\pi \|d_i\|^2 + \epsilon}, \quad (3)$$

where  $r_2 \sim U(0, 1)$ ,  $\epsilon$  is a small constant to avoid division by zero,  $d_i$  is the displacement between the current best solution and the  $i$ th agent,

$$d_i = X_{HBA,best} - X_{HBA,i}, \quad (4)$$

and  $S_i$  is a source-intensity term associated with neighboring agents. Following the original HBA rationale, it can be expressed as

$$S_i = \|X_{HBA,i} - X_{HBA,i+1}\|^2. \quad (5)$$

The adaptive parameter  $\alpha$  controls the transition from exploration to exploitation and decreases exponentially over iterations:

$$\alpha = C \exp\left(-\frac{t}{t_{max}}\right), \quad (6)$$

where  $C$  is a positive constant. In the MATLAB implementation of MPHBS,  $C = 2$ .

The search direction is controlled by a binary flag  $F \in \{-1, 1\}$ , selected at random:

$$F = \begin{cases} 1, & \text{with probability 0.5,} \\ -1, & \text{with probability 0.5.} \end{cases} \quad (7)$$

For compactness, let  $d_{ij} = X_{HBA,best,j} - X_{HBA,ij}$ .

Then, if a random number  $r < 0.5$ , HBA applies the digging-mode update:

$$X_{HBA,new,ij} = X_{HBA,best,j} + F\beta I_i X_{HBA,best,j} + Fr_3 \alpha d_{ij} \left| \cos(2\pi r_4) (1 - \cos(2\pi r_5)) \right|, \quad (8)$$

where  $\beta$  is a constant controlling the intensity contribution, and  $r_3, r_4, r_5 \sim U(0, 1)$ .

Otherwise, the algorithm applies the honey-mode update:

$$X_{HBA,new,ij} = X_{HBA,best,j} + Fr_7 \alpha d_{ij}, \quad (9)$$

where  $r_7 \sim U(0, 1)$ .

After generating a candidate position, each component is projected back into the feasible domain before evaluation:

$$X_{HBA,new,ij} \leftarrow \min(ub_j, \max(lb_j, X_{HBA,new,ij})). \quad (10)$$

The new candidate is accepted only if it improves the fitness of the current agent. Thus, the HBA population is updated through a greedy survival rule:

$$X_{HBA,i}^{t+1} = \begin{cases} X_{HBA,new,i}, & \text{if } f(X_{HBA,new,i}) < f(X_{HBA,i}^t), \\ X_{HBA,i}^t, & \text{otherwise.} \end{cases} \quad (11)$$

In parallel, the best solution of the HBA population is updated as

$$X_{HBA,best} = \arg \min_{i=1,\dots,N} f(X_{HBA,i}). \quad (12)$$

### 3.2. MPA

MPA models the interaction between predators and prey through a search process that alternates Brownian and Lévy movement patterns across three temporal phases [19]. In the implementation adopted in MPHBS, MPA evolves a population of candidate solutions and updates them according to stage-dependent motion rules, while preserving the best-so-far information through an elite-guided mechanism, a memory-saving step, and a final perturbation stage based on Fish Aggregating Devices (FADs).

The candidate population of MPA is represented by the matrix in Eq. (13), where  $N$  denotes the population size and  $Dim$  is the dimensionality of the optimization problem:

$$X_{MPA} = \begin{bmatrix} X_{MPA,11} & X_{MPA,12} & \cdots & X_{MPA,1Dim} \\ X_{MPA,21} & X_{MPA,22} & \cdots & X_{MPA,2Dim} \\ \vdots & \vdots & \ddots & \vdots \\ X_{MPA,N1} & X_{MPA,N2} & \cdots & X_{MPA,NDim} \end{bmatrix} \quad (13)$$

As in HBA, each row of  $X_{MPA}$  represents one  $Dim$ -dimensional candidate.

The initial positions are generated uniformly within the search bounds:

$$X_{MPA,ij} = lb_j + r_{ij}(ub_j - lb_j), \quad (14)$$

where  $r_{ij} \sim U(0, 1)$ , and  $lb_j$  and  $ub_j$  are the lower and upper bounds of the  $j$ th decision variable, respectively.

At each iteration, the current best solution found by MPA is replicated to form the Elite matrix:

$$Elite = \begin{bmatrix} X_{MPA,best} \\ X_{MPA,best} \\ \vdots \\ X_{MPA,best} \end{bmatrix} \in \mathbb{R}^{N \times Dim} \quad (15)$$

The algorithm also employs a time-varying control factor that regulates the step size during the search:

$$CF = \left(1 - \frac{t}{t_{max}}\right)^{\frac{2t}{t_{max}}} \quad (16)$$

where  $t$  is the current iteration and  $t_{max}$  is the maximum number of iterations.

In the first phase, corresponding to  $t < t_{max}/3$ , all agents are updated using Brownian-driven motion. Let  $RB_{ij}$  denote a Gaussian random variable and  $R_{ij} \sim U(0, 1)$ . The displacement term is computed as

$$\Delta X_{MPA,ij}^{(1)} = RB_{ij} (Elite_{ij} - RB_{ij} X_{MPA,ij}), \quad (17)$$

and the position update is given by

$$X_{MPA,ij}^{t+1} = X_{MPA,ij}^t + P R_{ij} \Delta X_{MPA,ij}^{(1)}, \quad (18)$$

where  $P = 0.5$  is a constant step-weight parameter.

In the second phase, corresponding to  $t_{max}/3 < t < 2t_{max}/3$ , the population is divided into two halves. For the first half of the population, Lévy-driven movement is used. Let  $RL_{ij}$  denote a Lévy-distributed random variable. The displacement is defined as

$$\Delta X_{MPA,ij}^{(2a)} = RL_{ij} (Elite_{ij} - RL_{ij} X_{MPA,ij}), \quad i = 1, \dots, \frac{N}{2}, \quad (19)$$

and the update is

$$X_{MPA,ij}^{t+1} = X_{MPA,ij}^t + P R_{ij} \Delta X_{MPA,ij}^{(2a)}, \quad i = 1, \dots, \frac{N}{2}. \quad (20)$$

For the second half of the population, Brownian motion is used around the elite solution:

$$\Delta X_{MPA,ij}^{(2b)} = RB_{ij} (RB_{ij} Elite_{ij} - X_{MPA,ij}), \quad i = \frac{N}{2} + 1, \dots, N, \quad (21)$$

and the update becomes

$$X_{MPA,ij}^{t+1} = Elite_{ij} + P CF \Delta X_{MPA,ij}^{(2b)}, \quad i = \frac{N}{2} + 1, \dots, N. \quad (22)$$

In the third phase, corresponding to  $t > 2t_{max}/3$ , all agents are updated through Lévy-driven motion around the elite solution:

$$\Delta X_{MPA,ij}^{(3)} = RL_{ij} (RL_{ij} Elite_{ij} - X_{MPA,ij}), \quad (23)$$

$$X_{MPA,ij}^{t+1} = Elite_{ij} + P CF \Delta X_{MPA,ij}^{(3)}. \quad (24)$$

After the stage-dependent movement update, all candidate positions are projected back into the feasible domain before fitness evaluation:

$$X_{MPA,ij} \leftarrow \min(ub_j, \max(lb_j, X_{MPA,ij})). \quad (25)$$

To preserve the best positions found so far, MPHBS adopts the memory-saving mechanism of MPA. Let  $f_{MPA}^{old}$  and  $X_{MPA}^{old}$  denote the fitness values and population stored from the previous memory state. The updated population is accepted only if it improves the stored solution; otherwise, the previous solution is retained. This operation can be written as

$$X_{MPA} \leftarrow \mathbb{I}(f_{MPA}^{old} < f_{MPA}) \odot X_{MPA}^{old} + \mathbb{I}(f_{MPA}^{old} \geq f_{MPA}) \odot X_{MPA}, \quad (26)$$

with the associated fitness update

$$f_{MPA} \leftarrow \mathbb{I}(f_{MPA}^{old} < f_{MPA}) \odot f_{MPA}^{old} + \mathbb{I}(f_{MPA}^{old} \geq f_{MPA}) \odot f_{MPA}, \quad (27)$$

where  $\mathbb{I}(\cdot)$  denotes an element-wise indicator mask and  $\odot$  is the Hadamard product.

Finally, MPA applies a perturbation stage controlled by FADs, which improves diversity and helps avoid premature convergence. Let  $U$  be a binary mask with entries sampled from a Bernoulli distribution with probability  $FADs$ , let  $r \sim U(0, 1)$ , and define the random bounded perturbation vector

$$Z = X_{min} + rand \odot (X_{max} - X_{min}). \quad (28)$$

The perturbation rule is then defined as

$$X_{MPA}^{new} = \begin{cases} X_{MPA} + CF Z \odot U & \text{if } r < FADs, \\ X_{MPA} + (FADs(1-r) + r) \times (X_{MPA}^{r_1} - X_{MPA}^{r_2}) & \text{otherwise,} \end{cases} \quad (29)$$

where  $X_{MPA}^{r_1}$  and  $X_{MPA}^{r_2}$  denote two random permutations of the population. The perturbed population then replaces the previous one:

$$X_{MPA} \leftarrow X_{MPA}^{new}. \quad (30)$$

### 3.3. SARSA algorithm

The SARSA algorithm is an on-policy RL method that estimates the action-value function  $Q(s, a)$  based on the agent's direct interaction with the environment. Its goal is to learn a policy that maximizes the cumulative reward by updating the state-action values according to the actions selected by the current policy.

SARSA stores action values in a Q-table and updates the state-action pair actually selected by its current policy. MPHBS later instantiates this on-policy memory over dimension, ranked candidate index, and source population.

The algorithm relies on the following update rule:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha [r_{t+1} + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)], \quad (31)$$

where  $s_t$  and  $a_t$  denote the state and action at time  $t$ ,  $r_{t+1}$  is the reward received after executing  $a_t$ ,  $s_{t+1}$  is the resulting next state,  $a_{t+1}$  is the subsequent action selected according to the same policy,  $\alpha \in (0, 1]$  is the learning rate, and  $\gamma \in [0, 1]$  is the discount factor for future rewards.

Action selection follows an  $\epsilon$ -greedy policy: a random feasible action is selected with probability  $\epsilon$ , and a maximizing action is selected otherwise.

The stochastic policy can be formally defined as follows:

$$\pi(a|s) = \begin{cases} 1 - \epsilon + \frac{\epsilon}{|\mathcal{A}(s)|}, & \text{if } a = \arg \max_{a'} Q(s, a'), \\ \frac{\epsilon}{|\mathcal{A}(s)|}, & \text{otherwise.} \end{cases} \quad (32)$$

where  $\mathcal{A}(s)$  is the set of possible actions in state  $s$ , and  $\epsilon \in [0, 1]$  controls the trade-off between exploration and exploitation. A high value of  $\epsilon$  promotes more exploration, while a low value encourages exploitation of the current policy.

The global best solution used by MPHBS is obtained by comparing the best solutions currently produced by HBA and MPA. In the following,  $P_g$  denotes the scalar global performance value, i.e., the best objective value found across both source optimizers, whereas  $X_{\text{best}}$  denotes the associated best decision vector. The subscript  $g$  indicates that this value is selected globally across the HBA and MPA populations. Accordingly, the global best update is defined as

$$(P_g, X_{\text{best}}) = \begin{cases} (f_{\text{MPA}, \text{best}}, X_{\text{MPA}, \text{best}}) & \text{if } f_{\text{MPA}, \text{best}} \leq f_{\text{HBA}, \text{best}}, \\ (f_{\text{HBA}, \text{best}}, X_{\text{HBA}, \text{best}}) & \text{otherwise.} \end{cases} \quad (33)$$

In MPHBS, the general on-policy update rule and  $\epsilon$ -greedy policy described above are instantiated over a tensorized state–action space indexed by problem dimension  $d$ , ranked candidate index  $n$ , and source population  $ch \in \{1, 2\}$ , as detailed in Section 4. The notation shifts from the generic pair  $(s_t, a_t)$  to the tuple  $(d, n, ch)$  in that section.

### 3.4. Structural and dynamic justification for the HBA–MPA pair

The choice of HBA and MPA as the two base optimizers in MPHBS is motivated by two concurrent properties: structural compatibility and dynamic complementarity. These two conditions are essential for the proposed mediator, since the SARSA-style information-exchange mechanism operates over aligned candidate memories and relies on the existence of heterogeneous search biases that can be exploited dimension-wise.

From a structural standpoint, both algorithms evolve continuous candidate populations of the form

$$X_{\text{HBA}}, X_{\text{MPA}} \in \mathbb{R}^{N \times \text{Dim}}, \quad (34)$$

where  $N$  denotes the population size and  $\text{Dim}$  is the dimensionality of the optimization problem. Eq. (34) implies that the candidate memories produced by both optimizers are directly alignable, which allows them to be concatenated and recombined dimension-wise without requiring encoding transformations, heterogeneous repair operators, or representation-specific mappings.

Their aligned representations permit ranked subsets from both populations to be organized as a source-indexed tensor and reused dimension-wise. The tensor construction and component-selection mechanism are defined once, together with the proposed framework, in Section 4.

Beyond representational compatibility, the HBA–MPA pair is also justified by the fact that the two algorithms induce different search dynamics. In compact form, the HBA update can be interpreted as

$$X_{\text{HBA}, i}^{t+1} = X_{\text{HBA}, \text{best}}^t + \Delta_{\text{HBA}, i}^t, \quad (35)$$

where  $\Delta_{\text{HBA}, i}^t$  is an intensity-guided displacement modulated by a decaying exploration factor. This produces a search process that is predominantly centered around the best solution found by HBA.

In contrast, the MPA dynamics are phase-dependent and can be written as

$$X_{\text{MPA}, i}^{t+1} = \Phi_t(X_{\text{MPA}, i}^t, X_{\text{MPA}, \text{best}}^t, \mathbf{R} \mathbf{B}_i^t, \mathbf{R} \mathbf{L}_i^t), \quad (36)$$

where  $\Phi_t(\cdot)$  alternates between Brownian-driven and Lévy-driven motion according to the current search phase. Consequently, MPA combines local refinement with heavy-tailed exploratory behavior, whereas HBA follows a more best-centered contraction dynamic.

Thus, HBA and MPA provide aligned but behaviorally non-redundant candidate memories: HBA contributes intensity-guided best-centered motion, whereas MPA contributes phase-dependent Brownian and Lévy dynamics. This complementarity is the relevant requirement for the mediator; equivalent source dynamics would offer little benefit for component reuse.

Dimension-wise reuse is most plausible when useful partial components persist in the ranked memories and cross-dimensional interactions do not dominate local behavior. This is a design rationale rather than a separability claim; strong variable coupling may reduce the benefit of independent component reuse, as acknowledged in the Threats to Validity section.

Therefore, the HBA–MPA pair is not selected arbitrarily. It satisfies the minimum structural requirements of the proposed mediator: common continuous representation, direct tensorizability, and heterogeneous search kernels. These properties make HBA and MPA a principled pair for a dimension-wise, memory-aware, SARSA-guided information-exchange strategy.

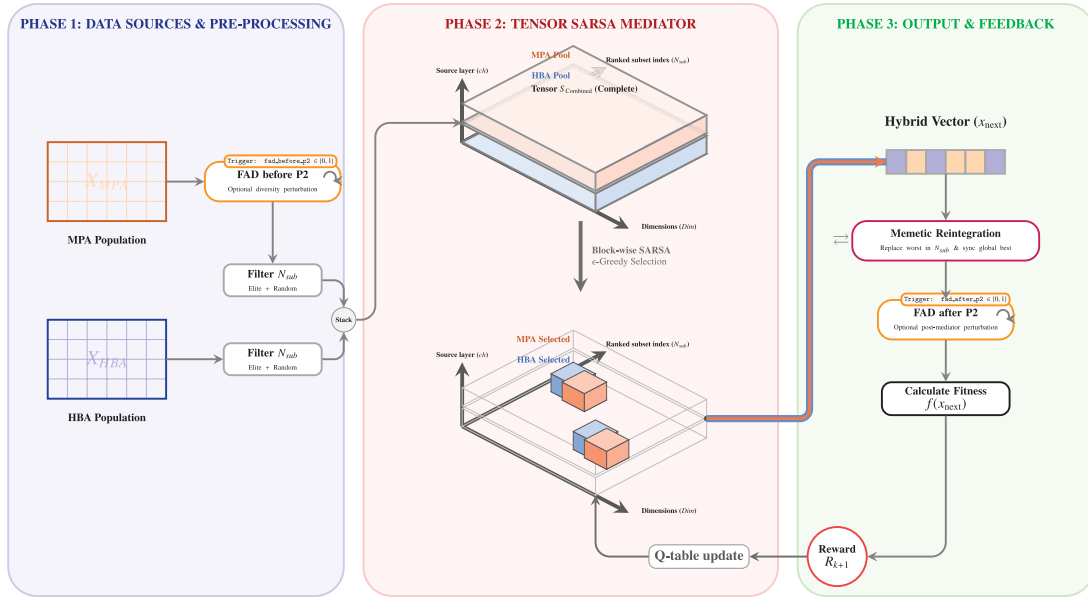
## 4. MPHBS framework

The proposed MPHBS framework integrates MPA and HBA through a SARSA-style reinforcement-learning mediator. Fig. 1 provides a conceptual overview of the data flow across three functional zones. The framework operates in two phases. In Phase 1, HBA and MPA are executed sequentially to generate updated candidate populations and their corresponding best solutions. In Phase 2, a SARSA-style mediator governs the information exchange by performing a fine-grained sub-exploration over selected components of both populations. Using Q-informed and exploratory action-selection behaviors, including an  $\epsilon$ -greedy policy, over a tensorized candidate representation, the mediator assembles hybrid solutions dimension by dimension, evaluates only one selected candidate per episode, and updates its decision values according to the improvement achieved with respect to the current assembled solution.

### 4.1. Phase 1: Sequential execution of algorithms

In Phase 1, HBA and MPA are executed sequentially within the same outer iteration to generate two heterogeneous candidate populations, denoted by  $X_{\text{HBA}}$  and  $X_{\text{MPA}}$ . Each optimizer updates its population according to its original search dynamics and maintains its own best solution throughout the process. For HBA, the position update is governed by Eqs. (8) and (9), whereas for MPA the update follows the stage-dependent rules defined by Eqs. (18), (20), (22), and (24), together with the elite-guided representation, the memory-saving mechanism, and the FADs perturbation stage.

The sequential execution is designed to preserve the native behavior of both optimizers before any reinforcement-learning-mediated exchange is applied. HBA contributes a best-centered search trajectory driven by intensity-guided movements and greedy replacement, while MPA contributes a phase-dependent search trajectory based on Brownian and Lévy motion, elite guidance, memory saving, and FADs-based perturbation. As a result, at the end of Phase 1, the algorithm has two



**Fig. 1.** Refined architecture of the proposed MPHBS algorithm. Phase 1 produces HBA and MPA candidate memories, while Phase 2 uses a tensor SARSA mediator to perform dimension-wise source-aware component selection, reward computation, and Q-table update. The FAD before P2 and FAD after P2 blocks represent the domain-dependent placement analyzed in the internal validation, and the memetic reintegration block synchronizes the selected hybrid candidate with the search memories.

independently updated populations that encode different search biases over the same continuous decision space.

After both populations have been updated and evaluated, MPHBS identifies the best solution currently available in each source optimizer. The HBA branch provides  $(f_{HBA,best}, X_{HBA,best})$ , whereas the MPA branch provides  $(f_{MPA,best}, X_{MPA,best})$ . These two local best solutions are then compared to update the global pair  $(P_g, X_{best})$ , where  $P_g$  is the scalar global performance value and  $X_{best}$  is the associated decision vector. This global pair is later used by the mediator both as a reference solution and as a synchronization signal between the two source memories.

The role of Phase 1 is therefore not merely to advance two optimizers independently, but to prepare structured and complementary candidate memories for the subsequent information-exchange stage. Since both HBA and MPA operate over matrices of size  $N \times Dim$ , their updated populations can be aligned directly by dimension. This alignment is essential for Phase 2, where selected rows from both populations are ranked, stacked, and transformed into a tensorized memory from which the SARSA-style mediator performs source-aware component selection.

This design also prevents the reinforcement-learning component from overriding the base search processes. Instead, the mediator acts only after HBA and MPA have produced their updated candidate memories. In other words, Phase 1 preserves the exploration and exploitation mechanisms of the parent optimizers, while Phase 2 determines how useful components from their resulting populations should be recombined. Algorithm 1 summarizes the overall workflow of MPHBS, while the information-exchange mechanism corresponding to Phase 2 is detailed in Algorithm 2.

#### 4.2. Phase 2: SARSA-based information exchange

The information exchange in MPHBS is mediated by a SARSA-style reinforcement-learning mechanism that performs a fine-grained sub-exploration of the candidate solutions generated by HBA and MPA. Instead of operating directly on the full populations, the mediator first extracts selected subsets from both optimizers and organizes them into a tensorized candidate structure. This representation allows the algorithm to decide, for each problem dimension, which candidate

component should be selected and from which source population it should be taken.

To preserve meaningful action semantics, the selected subsets are ordered according to fitness before being used by the mediator. As a result, lower candidate indices correspond to better-ranked solution components within the current subset. This rank stabilization allows the reinforcement-learning component to learn dimension-wise preferences over candidate quality and source population, rather than over arbitrary row positions.

Moreover, the mediator does not necessarily evaluate every internally generated candidate assembly. Instead, it can form several low-cost hybrid candidates using the ranked tensor representation, score them through an inexpensive proxy, and evaluate only one selected candidate with the objective function. This strategy increases the amount of candidate exploration performed per expensive fitness evaluation.

Let  $X_{HBA}^{sub}$  and  $X_{MPA}^{sub}$  denote the selected subsets extracted from the current HBA and MPA populations, respectively. In the conceptual formulation, both subsets have size  $N_{sub} \times Dim$ , where  $N_{sub}$  is the subset size and  $Dim$  is the dimensionality of the optimization problem. These subsets are concatenated along the third dimension to form a tensorized candidate structure:

$$S_{Combined} = \text{CONCAT}(X_{HBA}^{sub}, X_{MPA}^{sub}) \quad (37)$$

Thus,  $S_{Combined} \in \mathbb{R}^{N_{sub} \times Dim \times 2}$ , where the third dimension identifies the source of each candidate component. The first layer corresponds to HBA and the second layer corresponds to MPA, as illustrated in Fig. 2. For a given problem dimension  $d$ , the mediator can therefore select a component from a ranked candidate index  $n \in \{1, \dots, N_{sub}\}$  and a source indicator  $ch \in \{1, 2\}$ . In practice, this tensor acts as a structured candidate memory from which hybrid solutions are assembled component-wise, enabling component-wise hybrid-vector assembly through dimension-wise, source-aware component reuse.

The reinforcement-learning memory is represented by a Q-table whose entries quantify the utility of selecting a candidate component for a given problem dimension. Accordingly, the Q-table is indexed by tuples of the form  $(d, n, ch)$ , where  $d$  denotes the problem dimension,  $n$  denotes the ranked candidate index within the current subset, and

**Algorithm 1** High-level pseudocode of MPHBS.

**Require:** Population size  $N$ , maximum outer iterations  $t_{\max}$ , bounds  $LB, UB$ , dimension  $Dim$ , objective function  $f_{obj}$ , mediator parameters  $K, N_i, \rho_{sub}, w_{best}, \gamma$ , and FAD placement flags  $fad\_before\_p2, fad\_after\_p2$ .

**Ensure:** Best solution  $X_{best}$ , best fitness  $P_g$ , and convergence curve  $Conv$ .

- 1: Convert  $LB$  and  $UB$  into row-wise bound vectors.
- 2: Set HBA parameters  $\beta$  and  $C$ ; set MPA parameters  $FADs$  and  $P$ .
- 3: Set mediator constants  $\alpha_{lr}, w_{rank}, w_{novel}, Num\_Blocks$ , and  $N_{sub}$ .
- 4: Initialize HBA and MPA populations using Eqs. (1)–(2) and Eqs. (13)–(14).
- 5: Evaluate both populations and initialize  $X_{HBA, best}, X_{MPA, best}, f_{HBA, best}$ , and  $f_{MPA, best}$ .
- 6: Initialize the MPA memory variables  $X_{MPA}^{old}$  and  $f_{MPA}^{old}$ .
- 7: Initialize the persistent Q-table  $Q \in \mathbb{R}^{Dim \times N_{sub} \times 2}$  using Eq. (38).
- 8: Initialize the global best pair  $(P_g, X_{best})$  using Eq. (33).
- 9: **while**  $t < t_{\max}$  **do**
- 10: Compute HBA drivers using Eqs. (3)–(7), and MPA drivers using Eqs. (15)–(17), (19), (21), and (23).
- 11: **for**  $i = 1$  to  $N$  **do**
- 12: Update  $X_{HBA}(i, :)$  using the HBA movement rules in Eqs. (8) and (9).
- 13: Update  $X_{MPA}(i, :)$  using the MPA phase-dependent rules in Eqs. (18), (20), (22), and (24).
- 14: Repair bounds and apply HBA greedy replacement using Eqs. (10) and (11).
- 15: **end for**
- 16: Apply MPA bound handling using Eq. (25), evaluate  $X_{MPA}$ , and update  $X_{MPA, best}$  and  $f_{MPA, best}$ .
- 17: Apply the MPA memory-saving mechanism using Eqs. (26) and (27).
- 18: Update  $X_{HBA, best}$  and  $f_{HBA, best}$  using Eq. (12), and refresh the corresponding MPA best values.
- 19: **if**  $fad\_before\_p2$  **then**
- 20: Apply the MPA FADs perturbation stage before the mediator using Eqs. (28)–(30).
- 21: Re-evaluate perturbed MPA candidates and refresh  $X_{MPA, best}$ .
- 22: **end if**
- 23: Update the current global best  $(P_g, X_{best})$  using Eq. (33).
- 24: Apply the SARSA-style information-exchange mediator in Algorithm 2.
- 25: Receive updated  $X_{HBA}, X_{MPA}, Q$ , and  $(P_g, X_{best})$ .
- 26: Refresh  $X_{HBA, best}, X_{MPA, best}$ , and  $(P_g, X_{best})$  using Eq. (33).
- 27: **if**  $fad\_after\_p2$  **then**
- 28: Apply the post-mediator MPA FADs perturbation stage using Eqs. (28)–(30).
- 29: Re-evaluate the perturbed MPA candidates and update the global best if improved.
- 30: **end if**
- 31:  $t \leftarrow t + 1$ .
- 32:  $Conv(t) \leftarrow P_g$ .
- 33: **end while**
- 34: **return**  $X_{best}, P_g$ , and  $Conv$ .

$ch \in \{1, 2\}$  identifies the source population. The initialization is given by

$$Q = \text{zeros}(Dim, N_{sub}, 2) \quad (38)$$

where  $Q(d, n, ch)$  stores the estimated value of choosing candidate rank  $n$  from source  $ch$  for dimension  $d$ . In MPHBS, this Q-table is persistent across outer iterations, which allows the mediator to accumulate decision preferences throughout the search process instead of restarting its experience at each call.

For each active dimension  $d$ , the mediator selects an action defined as a tuple

$$a_d^{(e)} = (n_d^{(e)}, ch_d^{(e)}),$$

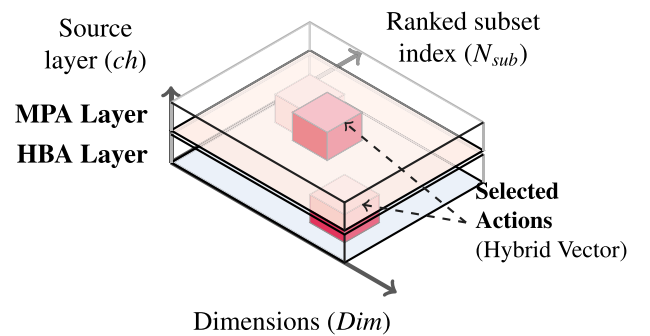
where  $n_d^{(e)}$  is the ranked candidate index selected for dimension  $d$  at episode  $e$ , and  $ch_d^{(e)}$  indicates whether the component is taken from the HBA layer or from the MPA layer. The action-selection process follows

**Algorithm 2** Subexploration of solutions using a SARSA-style mediator.

**Require:** Candidate matrices  $X_{HBA}, X_{MPA}$ ; their fitness vectors; persistent Q-table  $Q$ ; objective function  $f_{obj}$ ; current global best solution  $P_{best}$  and fitness  $F_{best}$ ; mediator parameters  $K, N_i, \rho_{sub}, \gamma, \alpha_{lr}, w_{rank}, w_{novel}, w_{best}$ , and  $Num\_Blocks$ .

**Ensure:** Updated candidate matrices  $X_{HBA}, X_{MPA}$ , global best pair  $(P_{best}, F_{best})$ , and Q-table  $Q$ .

- 1: Select elite-plus-random subsets from  $X_{HBA}$  and  $X_{MPA}$ , rank them by fitness, and construct  $S_{Combined}$  using Eq. (37).
- 2: Initialize  $x_{current} \leftarrow P_{best}$  and project it onto  $S_{Combined}$  with ClosestAction to obtain  $SA_{current}$ .
- 3:  $CurrentFitness \leftarrow f_{obj}(x_{current})$ .
- 4: Set  $N_{episodes} = N_{sub}N_i$ , define the active block size from  $Num\_Blocks$ , shuffle the dimension order, and use the persistent Q-table in Eq. (38).
- 5: **for**  $episode = 1$  to  $N_{episodes}$  **do**
- 6: Select the next block of active dimensions  $D_{act}$  from the shuffled dimension order; reshuffle when the pointer reaches the end.
- 7: **for**  $k = 1$  to  $K$  **do**
- 8: Generate  $x^{(k)}$  and  $SA^{(k)}$  over  $D_{act}$  using Eqs. (39)–(45).
- 9: Compute  $Score^{(k)}$  with the rank, novelty, and best-distance proxy using Eqs. (48)–(51).
- 10: **end for**
- 11: Select  $k^*$  using Eq. (52), set  $x_{next}$  using Eq. (53), and evaluate it once to obtain  $NextFitness$  using Eq. (54).
- 12: Compute  $Reward$  using Eq. (55).
- 13: Determine whether the transition is accepted using the relaxed acceptance rule in Eqs. (56)–(59).
- 14: **for**  $d \in D_{act}$  **do**
- 15: Update the selected entry of  $Q(d, n, ch)$  using Eqs. (60) and (61); accepted transitions bootstrap from the selected successor action, whereas rejected transitions bootstrap from the current action.
- 16: **end for**
- 17: **if** transition is accepted **then**
- 18: Set  $x_{current} \leftarrow x_{next}$ , update  $SA_{current}$ , replace the worst subset elements when improved, and rebuild  $S_{Combined}$ .
- 19: **end if**
- 20: **if**  $NextFitness < F_{best}$  **then**
- 21: Update  $(P_{best}, F_{best})$  and inject  $P_{best}$  into selected HBA and MPA positions.
- 22: **end if**
- 23: **end for**
- 24: Reintegrate the updated HBA and MPA subsets into their full populations and return.



**Fig. 2.** Representation of the tensorized candidate structure  $S_{Combined}$ . The diagram highlights selected components across problem dimensions, ranked subset indices  $N_{sub}$ , and source layers (HBA/MPA), illustrating the fine-grained source-aware sub-exploration mechanism.

an  $\epsilon$ -greedy policy with decay, so that the search gradually shifts from exploratory component selection to exploitation of previously rewarding choices.

Rather than modifying all dimensions simultaneously, the mediator operates on blocks of active dimensions. Let  $D_{act}^{(e)} \subseteq \{1, \dots, Dim\}$  denote the set of active dimensions at episode  $e$ . This blockwise strategy

reduces the combinatorial burden of full-vector reconstruction and allows the algorithm to perform localized solution refinement while preserving controlled exploration.

For one mediator call, the number of internal episodes is  $N_{episodes} = N_{sub} N_i$ , where  $N_i$  is the number of SARSA-style episodes assigned to each ranked subset element. This is the same quantity denoted by  $E$  in the computational-complexity analysis.

The implementation uses  $Num\_Blocks = 5$  and the fixed learning rate  $\alpha = \alpha_{lr} = 0.10$ . Accordingly, the active block size is  $B = \lceil Dim/Num\_Blocks \rceil$ ; when  $Dim < 10$ ,  $Num\_Blocks$  is set to one and the mediator activates the complete vector. These constants are held fixed in all sensitivity, ablation, and external-comparison experiments.

At each episode, the SARSA-style mediator generates  $K$  low-cost candidate assemblies over the active dimension block instead of directly evaluating a single successor solution. For each candidate  $k \in \{1, \dots, K\}$ , the corresponding hybrid vector is constructed by preserving inactive dimensions from  $x_{current}$  and replacing active dimensions with components from  $S_{Combined}$ :

$$x_d^{(k)} = \begin{cases} x_{current,d}, & d \notin D_{act}^{(e)}, \\ S_{Combined}(n_d^{(k)}, d, ch_d^{(k)}), & d \in D_{act}^{(e)}, \end{cases} \quad k = 1, \dots, K. \quad (39)$$

where the action for the  $d$ th active dimension in the  $k$ th candidate is defined as

$$a_d^{(k)} = (n_d^{(k)}, ch_d^{(k)}), \quad d \in D_{act}^{(e)}, \quad k = 1, \dots, K. \quad (40)$$

In the implemented version of MPHBS, the  $K$  candidate assemblies are generated from a small set of action-selection behaviors. When  $K < 5$ , only the first  $K$  behaviors are used; when  $K > 5$ , the additional candidates reuse the shifted  $\epsilon$ -greedy behavior with different random shifts. For each active dimension  $d$ , the associated action is sampled according to one of the following behaviors:

$$a_d^{(1)} = \arg \max_{n, ch} Q(d, n, ch), \quad (41)$$

$$a_d^{(2)} \sim \pi_{\epsilon(e)}(d), \quad (42)$$

$$n_d^{(3)} \sim \mathcal{U}\{1, \dots, N_{sub}\}, \quad ch_d^{(3)} \sim \mathcal{U}\{1, 2\}, \quad (43)$$

$$a_d^{(4)} = \text{SecondBest}_{n, ch} Q(d, n, ch), \quad (44)$$

$$a_d^{(5)} \sim \pi_{\epsilon(\tilde{e})}(d), \quad \tilde{e} = \min(N_{episodes}, e + \delta), \quad (45)$$

where

$$\epsilon(e) = \epsilon_{end} + (\epsilon_{start} - \epsilon_{end}) \exp\left(-\frac{e-1}{\tau_\epsilon}\right), \quad (46)$$

and

$$\delta \sim \mathcal{U}\left\{1, \dots, \max\left(1, \left\lfloor \frac{N_{episodes}}{10} \right\rfloor\right)\right\}. \quad (47)$$

Since evaluating all  $K$  candidate assemblies with the objective function would increase the computational budget, MPHBS assigns a low-cost proxy score to each candidate and evaluates only the highest-scoring one. The proxy combines three terms: candidate rank quality, local novelty with respect to the current assembled solution, and proximity to the current global best.

First, the average rank of candidate  $k$  over the active block is computed as

$$\bar{r}^{(k)} = \frac{1}{|D_{act}^{(e)}|} \sum_{d \in D_{act}^{(e)}} n_d^{(k)}. \quad (48)$$

Then, the novelty and best-distance terms are defined as

$$Novelty^{(k)} = \frac{1}{|D_{act}^{(e)}|} \sum_{d \in D_{act}^{(e)}} |x_d^{(k)} - x_{current,d}|, \quad (49)$$

$$Dist_{best}^{(k)} = \left\| x_{D_{act}^{(e)}}^{(k)} - P_{best, D_{act}^{(e)}} \right\|_2. \quad (50)$$

The final proxy score is given by

$$Score^{(k)} = -w_{rank} \bar{r}^{(k)} + w_{novel} Novelty^{(k)} - w_{best} Dist_{best}^{(k)}. \quad (51)$$

In all experiments, the proxy weights were fixed to  $w_{rank} = 1.00$  and  $w_{novel} = 0.30$ , whereas  $w_{best}$  was treated as part of the validated MPHBS configuration grid. Thus, the sensitivity analysis changes only the best-distance pressure in the proxy while keeping the rank-quality and local-novelty contributions constant across all configurations.

The candidate selected for expensive evaluation is the one with the highest proxy score:

$$k^* = \arg \max_{k \in \{1, \dots, K\}} Score^{(k)}. \quad (52)$$

It is important to clarify that this proxy is not intended to act as a surrogate objective function. Its role is limited to local screening among the hybrid candidates generated within the same episode. Final acceptance, reward computation, and global-best updates remain based on the true objective value. Therefore, the proxy does not require global calibration in the same sense as a fitted surrogate model; it only needs to provide a coarse preference signal before the expensive evaluation is performed. This conservative design keeps the mediator lightweight while limiting the effect of possible scale biases to the candidate-screening stage.

Thus, the next candidate to be evaluated is

$$x_{next} = x^{(k^*)}. \quad (53)$$

Although  $K$  candidate assemblies are internally generated at each episode, only one objective-function evaluation is performed:

$$F(x_{next}) = f_{obj}(x_{next}). \quad (54)$$

Therefore, MPHBS preserves a one-expensive-evaluation-per-episode budget while increasing the amount of candidate exploration carried out before committing to a transition.

After assembling the selected hybrid vector  $x_{next}$ , the objective function is evaluated once to obtain its fitness value. The reward is local with respect to the current assembled solution  $x_{current}$ , while the global best is updated separately whenever a superior solution is found. This separates the learning signal from the global-best replacement rule and provides a denser reward than one based exclusively on improvements over the global best. In the implemented version, the reward is ternary:

$$R_{k+1} = \begin{cases} 1, & F(x_{next}) < F(x_{current}), \\ 0, & F(x_{next}) = F(x_{current}), \\ -1, & F(x_{next}) > F(x_{current}). \end{cases} \quad (55)$$

Thus, the SARSA-style update receives a positive signal only when the evaluated hybrid candidate improves the current assembled solution, a neutral signal under equality, and a negative signal otherwise.

The transition from  $x_{current}$  to  $x_{next}$  follows the relaxed acceptance rule used in the implementation. Let  $t$  denote the current outer iteration,  $T_{max}$  the total number of outer iterations, and  $e$  the current mediator episode. For compactness, denote  $F_c = F(x_{current})$ ,  $F_n = F(x_{next})$ ,  $T_0 = \max(1, T_{max}-1)$ , and  $E_0 = \max(1, N_{episodes}-1)$ . The terms  $T_0$  and  $E_0$  prevent division by zero and make the global and local progress factors explicit:

$$p_g = \min\left(1, \frac{t}{T_0}\right), \quad p_\ell = \frac{e-1}{E_0}. \quad (56)$$

Here,  $p_g$  reduces exploratory acceptance as the outer search advances, whereas  $p_\ell$  plays the same role inside the mediator episode sequence. The relative deterioration of the evaluated candidate is measured with respect to the current assembled solution as

$$A_{rel} = \frac{F_n - F_c}{|F_c| + \epsilon_{mach}}. \quad (57)$$

where  $\epsilon_{\text{mach}}$  is machine precision. For a non-improving candidate, the acceptance probability is obtained by combining a progress-dependent decay factor with an exponential penalty on relative deterioration:

$$a_{\text{acc}} = 0.20(1 - p_g)^2 (0.35 + 0.65(1 - p_\ell)), \quad (58)$$

$$p_{\text{acc}} = \min\{0.25, \max[0, a_{\text{acc}} e^{-4 \max(0, A_{\text{rel}})}]\}.$$

The complete transition decision is therefore

$$\text{Accepted} = \begin{cases} 1, & F_n < F_c, \\ \mathbb{I}[u < p_{\text{acc}}], & F_n \geq F_c, \end{cases} \quad u \sim \mathcal{U}(0, 1). \quad (59)$$

Thus, all improving candidates are accepted, whereas non-improving candidates may be accepted with a probability that decreases with outer-iteration progress, local episode progress, and relative deterioration. The upper bound of 0.25 preserves occasional exploratory transitions without changing the fact that reward computation and global-best replacement are based on the true objective function.

The Q-table is updated after evaluating the selected hybrid candidate. Since the reward is obtained from the fitness of the complete hybrid vector, the same scalar reward is propagated to the dimension-wise decisions that participated in its construction. For a selected action  $(n, ch)$  at dimension  $d$ , the temporal-difference update is given by

$$Q(d, n, ch) \leftarrow Q(d, n, ch) + \alpha [R_{k+1} + \gamma \tilde{Q}_{\text{next}}(d) - Q(d, n, ch)]. \quad (60)$$

where

$$\tilde{Q}_{\text{next}}(d) = \begin{cases} Q(d, n', ch'), & \text{if the transition is accepted,} \\ Q(d, n, ch), & \text{otherwise.} \end{cases} \quad (61)$$

Here,  $(n', ch')$  denotes the action executed for the same dimension  $d$  in the selected successor candidate. The state index is therefore stationary at  $d$ : both the current value and the bootstrap term are evaluated in the same dimensional context, and no state transition is claimed in the sense of a sequential Markov decision process. More precisely, the mediator implements a dimension-wise contextual-bandit-style update with temporal-difference smoothing controlled by  $\gamma$ . The term SARSA-style is used in this restricted sense: the update retains the on-policy principle of bootstrapping from the action actually executed rather than from a maximization over all actions, without claiming the complete formal structure of a sequential MDP.

The  $\epsilon$ -greedy policy generates action choices within candidate assemblies, but it does not directly decide which candidate receives the expensive evaluation. That decision is made by the proxy in Eq. (52). Accordingly, the on-policy interpretation refers to the actions actually executed in the proxy-winning candidate, whose selected entries are propagated through Eqs. (60) and (61); rejected transitions retain the current action values in the bootstrap term.

In summary, the SARSA-style mediator in MPHBS does not replace the search dynamics of HBA or MPA; instead, it learns how to perform component-wise hybrid-vector assembly from ranked components generated by both optimizers. By combining blockwise decision-making, proxy-based candidate screening, and a single expensive evaluation per episode, the proposed mechanism increases the amount of useful exploration performed under a restricted function-evaluation budget.

#### 4.3. Computational complexity analysis

To evaluate the efficiency of the proposed MPHBS algorithm, we analyze its computational complexity in terms of time and space. The complexity depends on the population size  $N$ , the problem dimensionality  $D$ , the maximum number of outer iterations  $T_{\text{max}}$ , the subset ratio  $\rho_{\text{sub}}$ , the number of SARSA-style internal episodes per subset element  $N_i$ , the number of proxy candidates  $K$ , and the computational cost of evaluating the objective function, denoted by  $C_{\text{obj}}$ .

For clarity, let

$$N_{\text{sub}} = \max(5, \text{round}(\rho_{\text{sub}} N)), \quad (62)$$

and let the number of internal episodes in Phase 2 be

$$E = N_{\text{sub}} N_i. \quad (63)$$

In addition, let

$$B = \left\lceil \frac{D}{\text{Num\_Blocks}} \right\rceil \quad (64)$$

denote the size of the active dimension block used during the blockwise sub-exploration step.

The MPHBS framework consists of three main stages: initialization, the sequential execution of HBA and MPA (Phase 1), and the SARSA-style information exchange (Phase 2).

##### 4.3.1. Time complexity

**Initialization.** The initialization stage generates the HBA and MPA populations, each of size  $N \times D$ , and evaluates all initial candidate solutions. Therefore, the initialization cost is

$$\mathcal{O}(\text{Init}) = \mathcal{O}(2ND) + \mathcal{O}(2N C_{\text{obj}}) = \mathcal{O}(N(D + C_{\text{obj}})). \quad (65)$$

**Phase 1: Sequential execution of HBA and MPA.** In each outer iteration, HBA updates all  $N$  agents over  $D$  dimensions and evaluates one new candidate per agent. Hence, its per-iteration cost is

$$\mathcal{O}(\text{HBA}) = \mathcal{O}(ND) + \mathcal{O}(N C_{\text{obj}}). \quad (66)$$

Similarly, MPA performs stage-dependent updates over all  $N$  agents and  $D$  dimensions, followed by boundary handling and fitness evaluation. In addition, the implementation includes memory saving and a FADs perturbation stage, both of which are linear in the population matrix size. Thus, the per-iteration complexity of MPA is

$$\begin{aligned} \mathcal{O}(\text{MPA}) &= \mathcal{O}(ND) + \mathcal{O}(N C_{\text{obj}}) \\ &\quad + \mathcal{O}(ND) + \mathcal{O}(ND) \\ &= \mathcal{O}(N(D + C_{\text{obj}})). \end{aligned} \quad (67)$$

Therefore, the total complexity of Phase 1 per outer iteration is

$$\mathcal{O}(\text{Phase1}) = \mathcal{O}(\text{HBA}) + \mathcal{O}(\text{MPA}) = \mathcal{O}(N(D + C_{\text{obj}})). \quad (68)$$

More explicitly, if constant factors are retained, Phase 1 performs approximately  $2N$  objective-function evaluations per outer iteration, one for HBA and one for MPA per agent.

**Phase 2: SARSA-style information exchange.** The second phase contains four main components: subset extraction, ranking, candidate-tensor construction, and blockwise SARSA-style sub-exploration.

First, the mediator selects elite-plus-random subsets from both populations and ranks them according to fitness. Since the MATLAB implementation uses sorting operations, the ranking stage costs

$$\mathcal{O}(\text{Ranking}) = \mathcal{O}(N \log N). \quad (69)$$

Second, the combined tensor  $S_{\text{Combined}}$  is constructed from two subsets of size  $N_{\text{sub}} \times D$ , which requires

$$\mathcal{O}(\text{Tensor}) = \mathcal{O}(N_{\text{sub}} D). \quad (70)$$

Third, the initial assembled solution  $x_{\text{current}}$  is built by selecting one action per dimension, which requires

$$\mathcal{O}(\text{InitialState}) = \mathcal{O}(D) + \mathcal{O}(C_{\text{obj}}), \quad (71)$$

because the initial hybrid vector is evaluated once.

Fourth, the mediator runs  $E = N_{\text{sub}} N_i$  internal episodes. In each episode, it generates  $K$  low-cost candidate assemblies. In the actual

implementation, each candidate begins by copying the current full solution and its associated action records, which is linear in  $D$ . Therefore, the candidate-generation overhead per episode is

$$\mathcal{O}(\text{Candidates per episode}) = \mathcal{O}(KD). \quad (72)$$

The proxy score of each candidate is then computed over the active block, which requires rank aggregation, novelty computation, and distance-to-best evaluation. This contributes

$$\mathcal{O}(\text{Proxy per episode}) = \mathcal{O}(KB). \quad (73)$$

Since  $B \leq D$ , the candidate-generation and proxy-scoring costs together are bounded by  $\mathcal{O}(KD)$ .

Only one candidate, namely the proxy-best candidate, is evaluated with the objective function per episode. Hence, the expensive evaluation cost per episode is

$$\mathcal{O}(\text{Evaluation per episode}) = \mathcal{O}(C_{obj}). \quad (74)$$

Finally, the Q-table update is applied only over the active block, so its cost is

$$\mathcal{O}(\text{Q-update per episode}) = \mathcal{O}(B). \quad (75)$$

Combining these terms, the total cost per internal episode is

$$\mathcal{O}(\text{Episode}) = \mathcal{O}(KD + C_{obj} + B) = \mathcal{O}(KD + C_{obj}), \quad (76)$$

since  $B \leq D$ .

Therefore, the total complexity of Phase 2 per outer iteration is

$$\begin{aligned} \mathcal{O}(\text{Phase2}) &= \mathcal{O}(N \log N) + \mathcal{O}(N_{sub} D) \\ &\quad + \mathcal{O}(D + C_{obj}) + E \mathcal{O}(KD + C_{obj}), \end{aligned} \quad (77)$$

which can be written more compactly as

$$\mathcal{O}(\text{Phase2}) = \mathcal{O}(N \log N + N_{sub} D + E(KD + C_{obj})). \quad (78)$$

Using  $E = N_{sub} N_i$ , we obtain

$$\begin{aligned} \mathcal{O}(\text{Phase2}) &= \mathcal{O}(N \log N + N_{sub} D \\ &\quad + N_{sub} N_i (KD + C_{obj})). \end{aligned} \quad (79)$$

**Total time complexity.** The total computational cost of MPHBS over  $T_{max}$  outer iterations is therefore

$$\mathcal{O}(\text{MPHBS}) = \mathcal{O}(\text{Init}) + T_{max} [\mathcal{O}(\text{Phase1}) + \mathcal{O}(\text{Phase2})]. \quad (80)$$

Substituting the previous expressions yields

$$\begin{aligned} \mathcal{O}(\text{MPHBS}) &= \mathcal{O}(N(D + C_{obj})) \\ &\quad + T_{max} \left[ \mathcal{O}(N(D + C_{obj})) \right. \\ &\quad \left. + \mathcal{O}(N \log N + N_{sub} D \right. \\ &\quad \left. + N_{sub} N_i (KD + C_{obj})) \right]. \end{aligned} \quad (81)$$

Assuming  $\rho_{sub}$ ,  $N_i$ , and  $K$  are treated as bounded constants, then  $N_{sub} = \mathcal{O}(N)$ , and the total complexity simplifies to

$$\mathcal{O}(\text{MPHBS}) = \mathcal{O}(T_{max} [N \log N + ND + NC_{obj}]). \quad (82)$$

In many engineering optimization problems, the objective-function evaluation dominates the arithmetic overhead, that is,  $C_{obj} \gg D$  and  $C_{obj} \gg \log N$ . Under this practically relevant assumption, the effective time complexity becomes

$$\mathcal{O}(\text{MPHBS}) \approx \mathcal{O}(T_{max} NC_{obj}). \quad (83)$$

Thus, although the SARSA-style mediator introduces additional ranking, tensor-construction, and proxy-screening operations, the asymptotic dominant term remains linear in the population size and the number of outer iterations with respect to the objective-function cost.

**Objective-function evaluation budget.** From the standpoint of expensive objective-function calls, the implementation performs

$$FE_{\text{init}} = 2N \quad (84)$$

during initialization, and per outer iteration

$$FE_{\text{iter}} = 2N + 1 + E, \quad (85)$$

where  $2N$  correspond to the HBA and MPA evaluations in Phase 1, one evaluation is required for the initial assembled solution in Phase 2, and  $E = N_{sub} N_i$  evaluations correspond to the one-evaluation-per-episode rule of the SARSA-style mediator. Hence, the total number of expensive evaluations is

$$FE_{\text{total}} = 2N + T_{max}(2N + 1 + E). \quad (86)$$

This expression makes explicit that the proxy-based  $K$ -candidate mechanism increases candidate exploration without increasing the number of expensive objective evaluations beyond one per internal episode.

#### 4.3.2. Space complexity

The space complexity is determined by the storage required for the candidate populations, the auxiliary matrices used by HBA and MPA, and the reinforcement-learning structures used in the SARSA-style mediator.

**Population and optimizer storage.** The algorithm stores both HBA and MPA populations, each of size  $N \times D$ . In addition, the implementation uses several auxiliary matrices of the same order, including updated populations, old MPA memory, elite representations, Brownian and Lévy random matrices, and bound-related auxiliary structures. Since the number of such matrices is constant with respect to  $N$  and  $D$ , the total storage required by the metaheuristic phase is

$$\mathcal{O}(\text{Populations}) = \mathcal{O}(ND). \quad (87)$$

**Tensor and Q-table storage.** The selected subsets  $X_{HBA}^{sub}$  and  $X_{MPA}^{sub}$  have size  $N_{sub} \times D$ , and the tensorized candidate structure  $S_{\text{Combined}}$  has size  $N_{sub} \times D \times 2$ . Likewise, the Q-table has dimensions  $D \times N_{sub} \times 2$ . Therefore,

$$\mathcal{O}(\text{Tensor} + \text{Q}) = \mathcal{O}(N_{sub} D). \quad (88)$$

**Episode-level auxiliary storage.** During Phase 2, the implementation stores the current assembled solution, action records, and the  $K$  candidate vectors and their associated action traces. Since each candidate stores a full vector of size  $D$ , the corresponding auxiliary memory is

$$\mathcal{O}(\text{Candidates}) = \mathcal{O}(KD). \quad (89)$$

**Total space complexity.** Combining the previous terms yields

$$\mathcal{O}(\text{Space}) = \mathcal{O}(ND) + \mathcal{O}(N_{sub} D) + \mathcal{O}(KD). \quad (90)$$

If  $\rho_{sub}$  is constant, then  $N_{sub} = \mathcal{O}(N)$ . If  $K$  is also treated as a small constant, the overall space complexity simplifies to

$$\mathcal{O}(\text{Space}) = \mathcal{O}(ND). \quad (91)$$

Therefore, the proposed hybridization preserves linear memory growth with respect to the population size and problem dimensionality, while the additional reinforcement-learning structures only contribute lower-order terms when  $K$  and  $\rho_{sub}$  remain bounded.

## 5. Experimental results

This section presents the experimental assessment of MPHBS from three complementary perspectives. First, the benchmark domains and the experimental protocol used throughout the study are defined. Second, an internal validation stage is conducted to analyze the behavior of the proposed architecture through hyperparameter sensitivity analysis and ablation experiments on CEC2022 and FIR filter design problems. Third, the final version of MPHBS is compared against baseline and recent metaheuristics on external CEC2022 and FIR benchmark scenarios.

All experiments were implemented in MATLAB 2024b and executed on a Mac mini equipped with an Apple M4 Pro processor, 24 GB of RAM, and 512 GB of storage, running macOS 26.3.1. Unless otherwise stated, all compared methods were executed under matched stopping criteria and function-evaluation budgets in order to ensure a fair comparison. Each configuration was executed under a common seed schedule 30 times, and the reported results are based on mean and standard deviation statistics computed from these repeated runs. In addition, nonparametric statistical tests were employed in the comparative stages to determine whether the observed performance differences are statistically significant.

The final internal-validation configuration of MPHBS was selected through a structured sensitivity procedure. Rather than exhaustively exploring all parameters simultaneously, the search was progressively reduced to a small set of promising configurations. The final selection was based on a multi-criteria ranking strategy that prioritized average rank across benchmark functions or cases, followed by global mean performance, variability, and computational cost. Therefore, the selected setting should be interpreted as the best overall trade-off across the benchmark domain, rather than as the best isolated performance on a single test instance.

### 5.1. Benchmark problems and experimental protocol

The internal validation of the proposed architecture was conducted on two benchmark domains with complementary characteristics. First, the CEC2022 benchmark suite was used to analyze the sensitivity and ablation behavior of the method under a standard black-box continuous optimization setting. Second, an FIR filter design benchmark was employed to evaluate the same architecture on a structured signal-processing optimization problem. In both domains, all compared methods were executed under matched evaluation budgets and repeated-run protocols.

#### 5.1.1. CEC2022 benchmark setting

For the CEC2022 internal validation stage, all 12 benchmark functions were considered at dimension  $D = 20$ . Each method was executed under the common seed schedule 30 times, using a population size of  $N = 30$ , and the maximum computational budget was fixed to  $10,000 \times D$  function evaluations per run. The benchmark functions were evaluated through the official `cec22_test_func` interface. Objective calls were counted directly by the evaluation wrapper; observed consumption across the final CEC2022 campaigns ranged from 199,614 to 200,000 FEs, without exceeding the declared budget.

Let  $x = (x_1, \dots, x_D) \in \mathbb{R}^D$  denote a candidate solution. The feasible search domain is defined as

$$\Omega_{\text{CEC}} = \{ x \in \mathbb{R}^D : -100 \leq x_j \leq 100, \forall j = 1, \dots, D \}. \quad (92)$$

For each benchmark function  $f_i$ , the reported performance value corresponds to the residual error with respect to the known optimum  $f_i^*$ , namely

$$E_i(x) = f_i(x) - f_i^*. \quad (93)$$

Hence, lower values indicate better performance, and the ideal value is zero.

The CEC2022 sensitivity campaign was performed using the OLDF2 architecture with fixed mediator structure  $K = 7$  and  $N_i = 3$ . The hyperparameter exploration was carried out over four configurations:

$$C1 = (w_{\text{best}} = 0.050, \gamma = 0.30, \rho_{\text{sub}} = 0.25), \quad (94)$$

$$C2 = (w_{\text{best}} = 0.025, \gamma = 0.30, \rho_{\text{sub}} = 0.25), \quad (95)$$

$$C3 = (w_{\text{best}} = 0.050, \gamma = 0.20, \rho_{\text{sub}} = 0.25), \quad (96)$$

$$C4 = (w_{\text{best}} = 0.050, \gamma = 0.30, \rho_{\text{sub}} = 0.30), \quad (97)$$

each one evaluated under both BEFORE and AFTER FAD placements.

#### 5.1.2. FIR benchmark setting

The second benchmark domain used for internal validation corresponds to an FIR filter design problem. Unlike CEC2022, which represents a general-purpose black-box optimization scenario, the FIR benchmark evaluates the proposed architecture on a structured frequency-domain design task. Since this benchmark is not a standardized numerical suite such as CEC2022, its case definitions, spectral regions, and fitness construction are stated explicitly to ensure reproducibility and methodological transparency.

In all FIR experiments, the filter order was fixed to 30, so that the optimization dimension becomes

$$D = N_{\text{order}} + 1 = 31. \quad (98)$$

Accordingly, the decision vector is the filter coefficient vector

$$h = [h_0, h_1, \dots, h_{30}] \in \mathbb{R}^{31}, \quad (99)$$

subject to box constraints

$$-1 \leq h_j \leq 1, \quad j = 0, \dots, 30. \quad (100)$$

All FIR experiments were executed with population size  $N = 30$ , 30 runs under the common seed schedule, and a maximum budget of 150,000 function evaluations. Objective calls were counted directly, with observed consumption ranging from 149,718 to 150,000 FEs across the final FIR campaigns. The frequency response was evaluated using  $N_{\text{fft}} = 2048$ . Let

$$\omega_n = \frac{n}{N_{\text{fft}}/2}, \quad n = 0, 1, \dots, \frac{N_{\text{fft}}}{2}, \quad (101)$$

denote the normalized frequency grid over the non-redundant half-spectrum.

The benchmark consists of eight FIR cases, denoted FIR1–FIR8 to avoid ambiguity with the MPHBS configuration labels C1–C4:

- *FIR1*: LPF\_Easy, with passband edge  $F_p = 0.18$  and stopband edge  $F_s = 0.25$ .
- *FIR2*: LPF\_Hard, with passband edge  $F_p = 0.22$  and stopband edge  $F_s = 0.26$ .
- *FIR3*: HPF\_Easy, with stopband edge  $F_s = 0.75$  and passband edge  $F_p = 0.82$ .
- *FIR4*: HPF\_Hard, with stopband edge  $F_s = 0.74$  and passband edge  $F_p = 0.78$ .
- *FIR5*: BPF\_Wide, with  $(F_{s1}, F_{p1}, F_{s2}, F_{p2}) = (0.25, 0.35, 0.65, 0.75)$ .
- *FIR6*: BPF\_Narrow, with  $(F_{s1}, F_{p1}, F_{s2}, F_{p2}) = (0.32, 0.40, 0.60, 0.68)$ .
- *FIR7*: BSF\_Wide, with  $(F_{p1}, F_{s1}, F_{s2}, F_{p2}) = (0.25, 0.35, 0.65, 0.75)$ .
- *FIR8*: BSF\_Narrow, with  $(F_{p1}, F_{s1}, F_{s2}, F_{p2}) = (0.32, 0.40, 0.60, 0.68)$ .

For each case, the passband and stopband index sets are constructed from the corresponding spectral specification. For low-pass filters,

$$\mathcal{P} = \{n : \omega_n \leq F_p\}, \quad \mathcal{S} = \{n : \omega_n \geq F_s\}. \quad (102)$$

For high-pass filters,

$$\mathcal{P} = \{n : \omega_n \geq F_p\}, \quad \mathcal{S} = \{n : \omega_n \leq F_s\}. \quad (103)$$

For band-pass filters,

$$\begin{aligned} \mathcal{P} &= \{n : F_{p1} \leq \omega_n \leq F_{p2}\}, \\ \mathcal{S} &= \{n : \omega_n \leq F_{s1} \vee \omega_n \geq F_{s2}\}. \end{aligned} \quad (104)$$

For band-stop filters,

$$\begin{aligned} \mathcal{P} &= \{n : \omega_n \leq F_{p1} \vee \omega_n \geq F_{p2}\}, \\ \mathcal{S} &= \{n : F_{s1} \leq \omega_n \leq F_{s2}\}. \end{aligned} \quad (105)$$

Let  $H(\omega)$  denote the discrete Fourier transform of the candidate vector  $h$ , and let  $H_{half}$  be the associated non-redundant half-spectrum. The stopband error is computed as the mean squared magnitude of the response over the stopband region:

$$\text{MSE}_s = \frac{1}{|\mathcal{S}|} \sum_{n \in \mathcal{S}} (\Re\{H_{half}(n)\}^2 + \Im\{H_{half}(n)\}^2). \quad (106)$$

The passband error is defined as the mean squared deviation of the magnitude response from the ideal unit gain:

$$\text{MSE}_p = \frac{1}{|\mathcal{P}|} \sum_{n \in \mathcal{P}} (|H_{half}(n)| - 1)^2. \quad (107)$$

The scalar objective minimized by the metaheuristics is then

$$F(h) = W_p \text{MSE}_p + W_s \text{MSE}_s. \quad (108)$$

Therefore, the FIR benchmark is treated as a box-constrained optimization problem in which the spectral requirements are incorporated directly into the objective through a weighted penalty design, rather than through a separate feasibility-violation function. In the present study,  $W_p = 1$  was used in all cases, whereas  $W_s = 100$  was adopted for the Easy and Wide cases and  $W_s = 120$  for the Hard and Narrow cases. This choice penalizes stopband leakage more strongly than passband deviation, thereby promoting sharper spectral discrimination.

The FIR sensitivity campaign was carried out with fixed mediator structure  $K = 3$  and  $N_i = 9$ , using the same four hyperparameter configurations C1–C4 and the same BEFORE/AFTER FAD placements adopted in the CEC2022 campaign. This preserves a consistent experimental structure across both benchmark domains while still allowing the domain-dependent behavior of the proposed architecture to be examined.

## 5.2. Statistical analysis

The experimental comparison was designed to account for the stochastic nature of population-based metaheuristics. Each method was therefore executed over 30 stochastic runs for every function or FIR case using a common seed schedule, so run  $r$  of every method forms a paired observation within the same benchmark instance. The raw final values obtained in these runs were retained for statistical analysis. For CEC2022, the analyzed value corresponds to the residual error with respect to the known optimum of each function. For FIR, the analyzed value corresponds to the final scalar fitness defined by the weighted passband and stopband mean-square errors. In both cases, lower values indicate better performance.

Descriptive performance is reported through the mean and standard deviation computed over the 30 runs. The mean is used as the primary indicator of expected optimization quality, whereas the standard deviation provides a measure of run-to-run variability. In addition, function-wise or case-wise ranks are computed from the mean values, assigning rank one to the method with the lowest mean value in

each benchmark instance. The average rank is then used as a compact indicator of global behavior across all functions or cases. This ranking criterion is particularly useful because it reduces the influence of scale differences among benchmark functions and allows the relative position of each method to be compared across heterogeneous instances.

For the sensitivity analysis, the purpose of the ranking procedure is configurational selection rather than definitive hypothesis testing. The selected configuration is therefore the one that offers the best global trade-off across average rank, mean performance, variability, and computational cost. For the ablation and external comparison stages, the selected configuration is treated as the anchor method, and all other methods are compared against it using paired nonparametric statistics. This distinction avoids using statistical tests as a parameter-tuning mechanism while still providing inferential evidence for the final comparative claims.

Within each benchmark function or FIR case, anchor-versus-competitor comparisons are performed using the Wilcoxon signed-rank test on the 30 seed-matched run outcomes. This test was adopted because metaheuristic performance distributions are often non-Gaussian, skewed, or affected by outliers. For each instance-level comparison, the null hypothesis states that the median paired difference is zero. The resulting  $p$ -values are adjusted jointly with the Holm procedure over the complete anchor–competitor–instance family within each analysis stage and domain. The summary tables report how many significant instances favor the anchor or the competitor after this correction.

For the global multi-problem analysis, each CEC2022 function or FIR case is treated as one independent block. Friedman tests are applied to the mean-based ranks across these blocks, and Kendall's coefficient of concordance  $W$  is reported as the omnibus effect size. Values of  $W$  closer to one indicate stronger agreement in the relative ordering of the methods across instances. When the omnibus test is significant, anchor-based post hoc comparisons are obtained from the Friedman average-rank differences and corrected across competitors using Holm's procedure. This block-level analysis prevents the repeated runs from being treated as independent benchmark problems.

Two complementary W/T/L summaries are retained. The detailed transposed tables report run-level wins, ties, and losses within each function or FIR case from the perspective of the column method against the anchor. The compact global tables instead count benchmark instances from the anchor perspective, using the mean result of each method in each function or case. Neither summary replaces inferential testing: the run-level view describes local stochastic behavior, whereas the instance-level view shows whether the global rank result is broadly distributed across the benchmark. In the compact tables, “Sig. inst. A/C” gives the number of instance-level Wilcoxon comparisons that remain significant after domain-wide Holm correction in favor of the anchor or competitor, respectively.

The ablation stage also includes RANDOM mirrors of the selected SARSA configurations. These RANDOM counterparts preserve the same HBA–MPA backbone, ranked candidate memories, active dimension blocks, number of proxy candidates  $K$ , proxy-screening rule, and evaluation budget. The only change is that the SARSA action-selection mechanism is disabled: instead of selecting source-aware component actions through the Q-table and the  $\epsilon$ -greedy policy, the candidate rank and source layer are sampled uniformly at random from the same admissible tensor structure. Therefore, the RANDOM mirrors test whether the observed gains come from the learned information-exchange policy rather than merely from adding an extra recombination phase.

## 5.3. Computational cost and runtime reporting

Wall-clock time is reported as a secondary computational-cost indicator, whereas solution quality under matched function-evaluation budgets remains the primary comparison criterion. The purpose of this analysis is not to claim an asymptotic runtime advantage over every comparator, but to quantify the practical overhead introduced

**Table 2**

Mean wall-clock runtime recorded in the final external comparisons. Times are averaged over all benchmark instances and 30 runs per instance under the common seed schedule.

| Benchmark | Method    | Mean time (s) | Mean FEs |
|-----------|-----------|---------------|----------|
| CEC2022   | B-C1      | 1.983         | 199 930  |
| CEC2022   | MPHB      | 0.483         | 199 950  |
| CEC2022   | HBA       | 0.607         | 199 980  |
| CEC2022   | MPA       | 0.555         | 199 980  |
| CEC2022   | EODE      | 0.859         | 199 980  |
| CEC2022   | RLTLBO    | 0.356         | 199 950  |
| CEC2022   | QQLMPA    | 15.961        | 199 920  |
| CEC2022   | GWO       | 0.578         | 199 980  |
| CEC2022   | FDB-TLABC | 0.787         | 200 000  |
| CEC2022   | FTO       | 0.598         | 199 614  |
| FIR       | A-C1      | 3.123         | 149 857  |
| FIR       | MPHB      | 1.739         | 150 000  |
| FIR       | HBA       | 1.942         | 150 000  |
| FIR       | MPA       | 1.807         | 150 000  |
| FIR       | EODE      | 2.063         | 150 000  |
| FIR       | RLTLBO    | 1.546         | 150 000  |
| FIR       | QQLMPA    | 15.017        | 150 000  |
| FIR       | GWO       | 1.787         | 150 000  |
| FIR       | FDB-TLABC | 1.826         | 150 000  |
| FIR       | FTO       | 1.733         | 149 718  |

by the SARSA-style information-exchange mediator. This distinction is important because learning inside a metaheuristic is performed in a nonstationary search process: the population distribution, best solution, ranked memories, and useful actions change throughout the run. If the learning mechanism requires many additional objective-function calls or an expensive policy-training stage, its computational cost can dominate the optimizer.

In MPHBS, this risk is controlled by design. The learning component is tabular, dimension-wise, and restricted to ranked candidate memories; it does not train a deep model, does not replace the HBA or MPA operators, and does not evaluate all internally generated hybrid candidates. Instead, the mediator generates several low-cost candidates, screens them with a lightweight proxy, and evaluates only one candidate with the true objective function per episode. Therefore, the additional runtime reflects a controlled information-exchange overhead rather than an unconstrained learning phase.

Table 2 summarizes the mean wall-clock times recorded from the final external-comparison outputs. As expected, MPHBS is slower than MPHB and the parent optimizers because it performs the additional SARSA-guided mediator stage. However, the overhead remains practical under the adopted budgets. Moreover, compared with QQLMPA, which is also a learning-based MPA-related method, MPHBS is substantially faster while achieving stronger overall ranks in both benchmark domains. This supports the claim that effective learning-based hybridization should control the cost of online learning, rather than simply adding a learning mechanism to a population-based optimizer.

#### 5.4. Internal validation of the proposed architecture

The internal validation was designed to determine whether the behavior of MPHBS remains stable across different benchmark domains and whether its main structural choices are consistently beneficial. Two complementary settings were considered: the CEC2022 benchmark suite, representing a standard black-box continuous optimization scenario, and the FIR filter design benchmark, representing a structured signal-processing optimization problem. The sensitivity results reveal a clear domain-dependent pattern. In particular, the preferred FAD placement is not the same in both benchmarks, which indicates that the interaction between the SARSA-based mediator and the search dynamics depends on the problem landscape. Therefore, the internal validation is not only used to select promising hyperparameter configurations, but also to understand how the proposed architecture behaves under different optimization regimes.

##### 5.4.1. CEC2022 hyperparameter sensitivity analysis

Table 3 reports the sensitivity results of the SARSA-based configurations on CEC2022 under closely matched observed FE consumption. A first clear observation is that the BEFORE configurations consistently dominate the AFTER configurations in this domain. The four best SARSA-based average ranks are occupied by B-C1, B-C3, B-C4, and B-C2, with average ranks of 3.167, 3.917, 4.333, and 4.667, respectively, whereas the AFTER configurations obtain substantially weaker average ranks between 10.083 and 12.917. This separation indicates that, for CEC2022, applying the FAD mechanism before the mediator phase is considerably more compatible with the search dynamics induced by the SARSA-guided recombination.

The omnibus Friedman analysis confirms that the CEC2022 sensitivity configurations do not share the same rank behavior ( $p = 1.137 \times 10^{-10}$ ). Kendall's  $W = 0.421$  indicates a moderate-to-large global effect: the ordering is systematic enough to support configuration selection, while still reflecting substantial heterogeneity among the 12 benchmark functions.

A more detailed inspection at the function level confirms this trend. The BEFORE configurations provide the best SARSA-based average ranks and attain the lowest mean error on several functions, while MPHB remains particularly strong on a subset of the benchmark. Within the SARSA family, B-C1 provides both the best average rank and the best Friedman mean rank, with B-C3 remaining close in the global ordering. This suggests that the BEFORE family is robust as a whole, although B-C1 offers the best rank-based trade-off within the explored grid.

From a hyperparameter perspective, the results indicate that the combination  $w_{best} = 0.050$ ,  $\gamma = 0.30$ , and  $\rho_{sub} = 0.25$  provides the most favorable rank-based trade-off when the FAD mechanism is applied before the mediator. B-C3 remains competitive, but reducing  $\gamma$ , reducing  $w_{best}$ , or increasing  $\rho_{sub}$  does not improve the global rank ordering. Therefore, within the analyzed CEC2022 search space, B-C1 is selected as the reference sensitivity configuration for the subsequent internal analysis.

It is also worth noting that the ranks reported in Table 3 are global ranks computed over the full set of evaluated methods, not only over the subset displayed in the SARSA panel. This explains why some visible methods receive rank values close to 15–17 even though the table itself shows fewer columns. Fig. 3 complements this view by showing the convergence behavior of the internal-validation configurations under the same evaluation budget.

##### 5.4.2. FIR hyperparameter sensitivity analysis

Table 4 reports the sensitivity results of the SARSA-based configurations on the FIR benchmark. In contrast to the behavior observed on CEC2022, the dominant pattern in this domain is associated with the AFTER placement. Among the SARSA configurations, A-C1 obtains the best Friedman mean rank and the lowest global mean fitness, while A-C3 and the remaining AFTER settings also retain competitive behavior. By comparison, the BEFORE configurations occupy weaker positions in the global ordering. This result indicates that, in the FIR design domain, placing the FAD mechanism after the mediator is more compatible with the search dynamics than applying it before the mediator phase. RANDOM configurations are retained as controls for the subsequent ablation and are not treated as candidate final MPHBS settings.

The corresponding Friedman test is also significant for FIR ( $p = 8.072 \times 10^{-11}$ ), with Kendall's  $W = 0.639$ . This large concordance effect indicates that the rank separation between the AFTER and BEFORE families is comparatively consistent across the eight FIR cases, supporting the selection of A-C1 without treating the sensitivity stage as a post hoc superiority test.

A case-wise inspection further reinforces this conclusion. The lowest mean fitness values among the SARSA configurations are concentrated in the AFTER family, and A-C1 provides the best global balance according to the selection criteria. Therefore, unlike CEC2022, where

**Table 3**

CEC2022 sensitivity results for SARSA configurations (BEFORE + AFTER) with MPHBS as reference. For each function, the error mean, standard deviation, and global rank are reported in separate rows. Global ranks are computed from the full method set using mean values. Lower values are better. The minimum mean within each displayed panel is highlighted in bold.

| Function | Statistic | B-C1             | B-C2             | B-C3      | B-C4             | A-C1             | A-C2      | A-C3             | A-C4      | MPHBS            |
|----------|-----------|------------------|------------------|-----------|------------------|------------------|-----------|------------------|-----------|------------------|
| F1       | Mean      | 3.615e-07        | 2.845e-07        | 3.099e-07 | 2.608e-07        | <b>1.677e-09</b> | 1.568e-08 | 5.041e-09        | 2.925e-06 | 3.869e-04        |
|          | Std       | 2.687e-07        | 2.784e-07        | 2.517e-07 | 1.921e-07        | 8.341e-09        | 8.413e-08 | 2.745e-08        | 1.586e-05 | 3.925e-04        |
|          | Rank      | 11.000           | 9.000            | 10.000    | 8.000            | 3.000            | 6.000     | 5.000            | 13.000    | 14.000           |
| F2       | Mean      | 2.752e+01        | 3.446e+01        | 3.162e+01 | 3.071e+01        | 2.795e+01        | 2.985e+01 | <b>2.700e+01</b> | 3.425e+01 | 3.043e+01        |
|          | Std       | 2.399e+01        | 2.244e+01        | 2.434e+01 | 2.200e+01        | 2.293e+01        | 2.285e+01 | 2.320e+01        | 2.031e+01 | 2.219e+01        |
|          | Rank      | 2.000            | 13.000           | 8.000     | 7.000            | 3.000            | 5.000     | 1.000            | 12.000    | 6.000            |
| F3       | Mean      | <b>1.803e-05</b> | 1.917e-03        | 9.264e-04 | 7.189e-03        | 7.297e-01        | 7.103e-01 | 7.931e-01        | 1.123e+00 | 4.295e-02        |
|          | Std       | 2.712e-05        | 1.043e-02        | 4.979e-03 | 3.449e-02        | 7.776e-01        | 7.800e-01 | 9.031e-01        | 1.389e+00 | 8.786e-02        |
|          | Rank      | 1.000            | 3.000            | 2.000     | 4.000            | 7.000            | 6.000     | 8.000            | 15.000    | 5.000            |
| F4       | Mean      | 3.735e+01        | <b>3.107e+01</b> | 3.632e+01 | 3.592e+01        | 5.118e+01        | 4.633e+01 | 5.197e+01        | 5.391e+01 | 3.598e+01        |
|          | Std       | 1.087e+01        | 9.953e+00        | 1.042e+01 | 9.139e+00        | 1.417e+01        | 1.301e+01 | 1.614e+01        | 1.253e+01 | 8.547e+00        |
|          | Rank      | 5.000            | 1.000            | 4.000     | 2.000            | 15.000           | 13.000    | 16.000           | 17.000    | 3.000            |
| F5       | Mean      | 3.031e+01        | 4.425e+01        | 3.707e+01 | 3.884e+01        | 5.505e+02        | 4.796e+02 | 5.492e+02        | 5.667e+02 | <b>2.096e+01</b> |
|          | Std       | 3.510e+01        | 4.817e+01        | 5.048e+01 | 4.280e+01        | 2.637e+02        | 2.226e+02 | 2.686e+02        | 2.285e+02 | 2.341e+01        |
|          | Rank      | 2.000            | 5.000            | 3.000     | 4.000            | 16.000           | 14.000    | 15.000           | 17.000    | 1.000            |
| F6       | Mean      | 2.264e+01        | 2.775e+01        | 2.400e+01 | 2.802e+01        | 5.224e+01        | 5.771e+01 | 6.395e+01        | 5.111e+01 | <b>1.635e+01</b> |
|          | Std       | 1.691e+01        | 1.859e+01        | 2.025e+01 | 2.284e+01        | 3.304e+01        | 3.855e+01 | 3.630e+01        | 3.149e+01 | 1.075e+01        |
|          | Rank      | 2.000            | 4.000            | 3.000     | 5.000            | 8.000            | 14.000    | 15.000           | 6.000     | 1.000            |
| F7       | Mean      | 2.945e+01        | 3.378e+01        | 3.157e+01 | 3.366e+01        | 6.652e+01        | 6.193e+01 | 6.140e+01        | 7.412e+01 | <b>2.881e+01</b> |
|          | Std       | 1.045e+01        | 1.067e+01        | 9.975e+00 | 1.036e+01        | 2.912e+01        | 2.730e+01 | 2.588e+01        | 3.971e+01 | 5.945e+00        |
|          | Rank      | 2.000            | 5.000            | 3.000     | 4.000            | 12.000           | 11.000    | 9.000            | 17.000    | 1.000            |
| F8       | Mean      | 2.056e+01        | <b>2.016e+01</b> | 2.102e+01 | 2.109e+01        | 2.840e+01        | 3.503e+01 | 2.882e+01        | 3.236e+01 | 2.114e+01        |
|          | Std       | 2.297e+00        | 3.097e+00        | 4.168e-01 | 4.335e-01        | 2.221e+01        | 2.983e+01 | 2.246e+01        | 2.413e+01 | 5.232e-01        |
|          | Rank      | 2.000            | 1.000            | 3.000     | 4.000            | 8.000            | 15.000    | 9.000            | 11.000    | 5.000            |
| F9       | Mean      | 1.808e+02        | 1.808e+02        | 1.808e+02 | <b>1.808e+02</b> | 1.808e+02        | 1.808e+02 | 1.808e+02        | 1.808e+02 | 1.808e+02        |
|          | Std       | 1.049e-08        | 2.046e-09        | 7.831e-09 | 1.269e-09        | 5.414e-08        | 8.717e-07 | 2.402e-07        | 9.569e-07 | 3.823e-06        |
|          | Rank      | 4.000            | 2.000            | 3.000     | 1.000            | 5.000            | 11.000    | 9.000            | 12.000    | 17.000           |
| F10      | Mean      | <b>5.154e+01</b> | 8.170e+01        | 6.044e+01 | 9.759e+01        | 1.618e+02        | 1.537e+02 | 1.589e+02        | 1.929e+02 | 1.005e+02        |
|          | Std       | 5.894e+01        | 6.876e+01        | 5.820e+01 | 7.116e+01        | 1.008e+02        | 1.174e+02 | 1.123e+02        | 1.340e+02 | 9.948e-02        |
|          | Rank      | 1.000            | 3.000            | 2.000     | 4.000            | 11.000           | 7.000     | 8.000            | 17.000    | 5.000            |
| F11      | Mean      | 3.000e+02        | 3.033e+02        | 3.000e+02 | 3.200e+02        | 3.307e+02        | 3.307e+02 | 3.267e+02        | 3.033e+02 | <b>2.800e+02</b> |
|          | Std       | 6.433e+01        | 6.687e+01        | 6.433e+01 | 4.068e+01        | 4.630e+01        | 4.630e+01 | 4.498e+01        | 6.687e+01 | 1.186e+02        |
|          | Rank      | 2.000            | 5.000            | 3.000     | 8.000            | 16.000           | 17.000    | 13.000           | 4.000     | 1.000            |
| F12      | Mean      | 2.410e+02        | 2.412e+02        | 2.407e+02 | <b>2.387e+02</b> | 2.630e+02        | 2.596e+02 | 2.617e+02        | 2.596e+02 | 2.391e+02        |
|          | Std       | 7.529e+00        | 6.937e+00        | 4.523e+00 | 4.137e+00        | 1.732e+01        | 1.406e+01 | 1.742e+01        | 1.463e+01 | 4.469e+00        |
|          | Rank      | 4.000            | 5.000            | 3.000     | 1.000            | 17.000           | 15.000    | 16.000           | 14.000    | 2.000            |
| AvgRank  | Rank      | <b>3.167</b>     | 4.667            | 3.917     | 4.333            | 10.083           | 11.167    | 10.333           | 12.917    | 5.083            |

the BEFORE family is clearly preferable, the FIR benchmark favors the AFTER family, suggesting that the relative effectiveness of the mediator depends on the structure of the optimization landscape rather than on a universally dominant FAD placement.

From a hyperparameter perspective, the results indicate that the combination  $w_{best} = 0.050$ ,  $\gamma = 0.30$ , and  $\rho_{sub} = 0.25$  provides the best overall SARSA trade-off when combined with the AFTER placement. In particular, A-C1 obtains the best Friedman mean rank among the SARSA settings and the lowest global mean fitness. The remaining AFTER configurations continue to perform competitively, showing that the selected behavior is not restricted to a single isolated setting; however, none improves the global selection criteria achieved by A-C1. Consequently, A-C1 is selected as the reference sensitivity configuration for the subsequent FIR ablation analysis.

As in the CEC2022 sensitivity analysis, the reported ranks are global ranks computed over the full set of evaluated methods, rather than over the subset of methods displayed in the SARSA panel. This explains why some visible methods receive rank values close to the lower end of the global ordering even though the table itself reports only the SARSA configurations and the MPHBS baseline. Fig. 4 reports the corresponding convergence profiles for the FIR internal-validation stage.

#### 5.4.3. CEC2022 ablation study

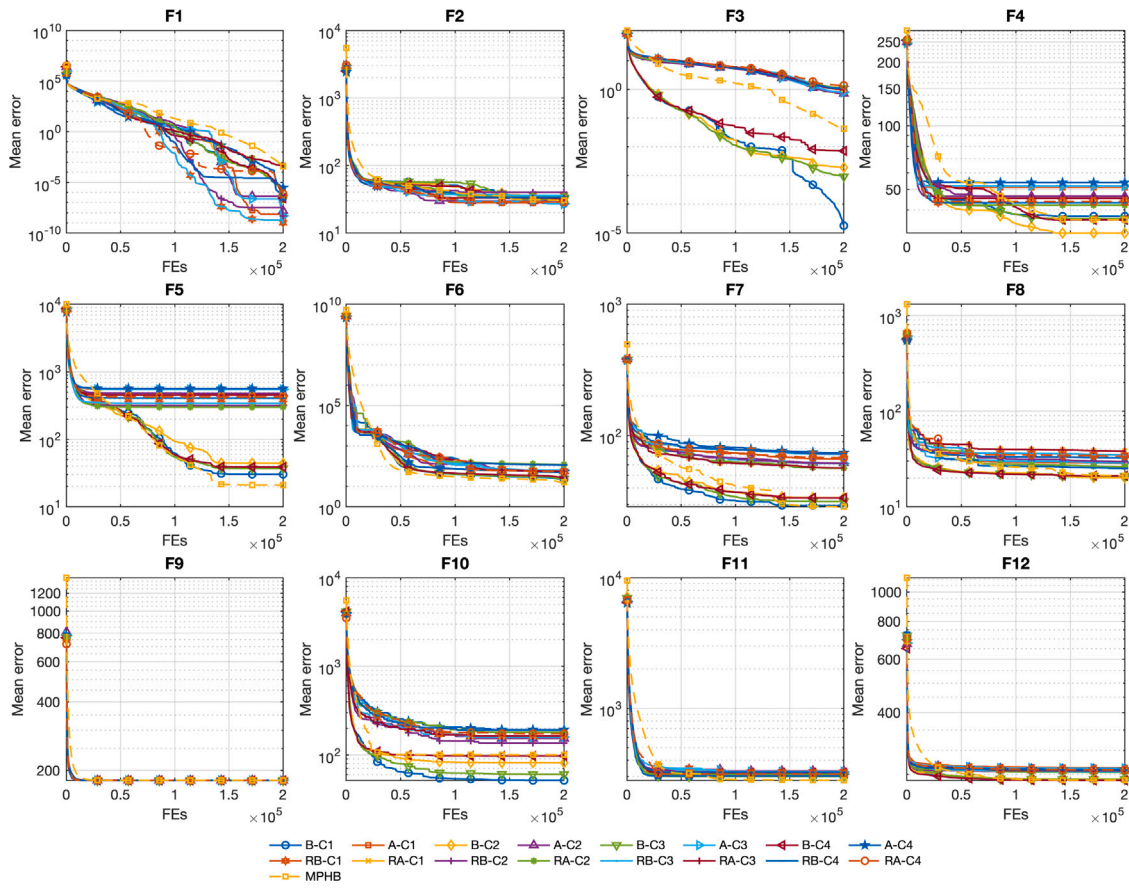
For consistency with the internal ablation tables, BS-C1 and AS-C1 denote the SARSA instances corresponding to the final B-C1 and A-C1 configurations, respectively. The prefixes only identify the SARSA

branch within the internal ablation files and do not define separate algorithms.

The CEC2022 ablation study was conducted using the final B-C1 configuration of MPHBS, reported as BS-C1 in the ablation tables, which was identified in the previous sensitivity analysis as the best-ranked SARSA configuration for this benchmark. The goal of this stage is to isolate the contribution of the SARSA-based mediator from the contribution of the remaining architectural components. To this end, the selected anchor was compared against its matched RANDOM mirror, the best RANDOM configuration identified in the same domain, the MPHBS baseline, and the two parent optimizers HBA and MPA.

A first important result is that B-C1 attains the best overall average rank among all methods included in the CEC2022 ablation. The detailed function-wise results are reported in Table 5. Although the anchor is not the best performer on every individual function, it provides the best global balance across the full benchmark set. B-C1 achieves the best mean error in F2, F3, F9, and F10; MPHBS and MPA lead different subsets of functions, and the matched RANDOM mirror leads F1. Therefore, the selected SARSA-based configuration is not uniformly dominant at the function level, but it is the most balanced overall method in the ablation set.

The clearest evidence in favor of the SARSA-based mediator emerges from the comparison against the RANDOM counterparts. At the function-block level, B-C1 records 11 wins and one loss against both its matched RANDOM mirror and R-best. The Friedman–Holm post hoc

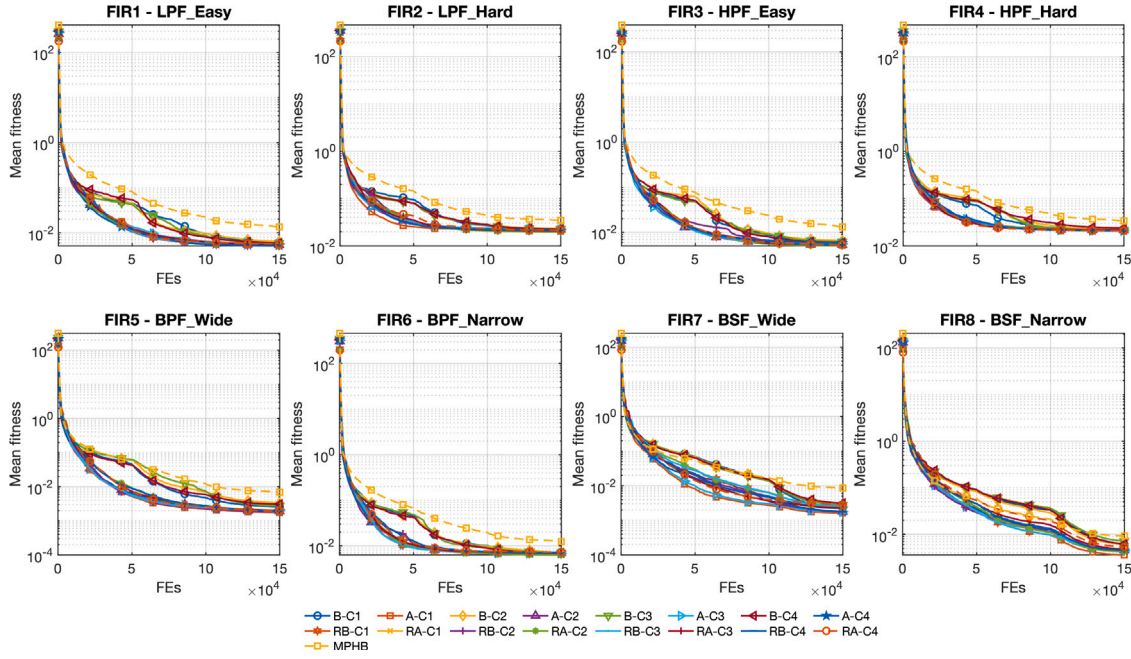


**Fig. 3.** Mean convergence curves on the 12 CEC2022 benchmark functions for the internal-validation configurations and the MPHB baseline. The horizontal axis reports function evaluations, and the vertical axis reports residual error in logarithmic scale. Each curve corresponds to the average of 30 paired runs.

**Table 4**

FIR sensitivity results for SARSA configurations (BEFORE + AFTER) with MPHB as reference. For each case, the fitness mean, standard deviation, and global rank are reported in separate rows. Global ranks are computed from the full method set using mean values. Lower values are better. The minimum mean within each displayed panel is highlighted in bold.

| Case    | Statistic | B-C1      | B-C2      | B-C3      | B-C4      | A-C1             | A-C2      | A-C3             | A-C4             | MPHB      |
|---------|-----------|-----------|-----------|-----------|-----------|------------------|-----------|------------------|------------------|-----------|
| FIR1    | Mean      | 6.178e-03 | 6.482e-03 | 5.886e-03 | 5.913e-03 | 5.739e-03        | 5.423e-03 | 5.608e-03        | <b>5.238e-03</b> | 1.356e-02 |
|         | Std       | 1.450e-03 | 1.413e-03 | 1.381e-03 | 1.703e-03 | 1.292e-03        | 7.602e-04 | 1.292e-03        | 7.887e-04        | 7.689e-03 |
|         | Rank      | 15.000    | 16.000    | 13.000    | 14.000    | 12.000           | 7.000     | 10.000           | 5.000            | 17.000    |
| FIR2    | Mean      | 2.298e-02 | 2.318e-02 | 2.266e-02 | 2.267e-02 | 2.147e-02        | 2.239e-02 | 2.204e-02        | <b>2.136e-02</b> | 3.448e-02 |
|         | Std       | 2.671e-03 | 2.679e-03 | 2.450e-03 | 2.499e-03 | 2.619e-03        | 2.352e-03 | 2.705e-03        | 1.973e-03        | 1.075e-02 |
|         | Rank      | 15.000    | 16.000    | 13.000    | 14.000    | 9.000            | 12.000    | 11.000           | 5.000            | 17.000    |
| FIR3    | Mean      | 5.728e-03 | 6.610e-03 | 6.233e-03 | 5.984e-03 | <b>5.157e-03</b> | 5.341e-03 | 5.583e-03        | 5.839e-03        | 1.333e-02 |
|         | Std       | 1.062e-03 | 1.921e-03 | 1.492e-03 | 1.420e-03 | 6.868e-04        | 6.461e-04 | 1.097e-03        | 1.383e-03        | 5.430e-03 |
|         | Rank      | 12.000    | 16.000    | 15.000    | 14.000    | 2.000            | 7.000     | 11.000           | 13.000           | 17.000    |
| FIR4    | Mean      | 2.181e-02 | 2.257e-02 | 2.166e-02 | 2.400e-02 | 2.149e-02        | 2.148e-02 | <b>2.084e-02</b> | 2.109e-02        | 3.379e-02 |
|         | Std       | 2.580e-03 | 2.573e-03 | 2.002e-03 | 4.152e-03 | 2.036e-03        | 1.869e-03 | 1.913e-03        | 2.256e-03        | 1.113e-02 |
|         | Rank      | 13.000    | 15.000    | 10.000    | 16.000    | 9.000            | 8.000     | 2.000            | 4.000            | 17.000    |
| FIR5    | Mean      | 2.597e-03 | 3.410e-03 | 2.757e-03 | 3.073e-03 | <b>1.760e-03</b> | 1.946e-03 | 1.993e-03        | 1.978e-03        | 6.874e-03 |
|         | Std       | 1.218e-03 | 2.900e-03 | 1.588e-03 | 1.627e-03 | 6.678e-04        | 8.545e-04 | 9.314e-04        | 7.770e-04        | 3.863e-03 |
|         | Rank      | 13.000    | 16.000    | 14.000    | 15.000    | 1.000            | 5.000     | 8.000            | 7.000            | 17.000    |
| FIR6    | Mean      | 6.877e-03 | 7.165e-03 | 7.130e-03 | 7.038e-03 | <b>6.668e-03</b> | 7.082e-03 | 6.826e-03        | 6.951e-03        | 1.235e-02 |
|         | Std       | 1.787e-03 | 2.522e-03 | 1.913e-03 | 1.771e-03 | 1.672e-03        | 1.692e-03 | 1.798e-03        | 1.662e-03        | 5.179e-03 |
|         | Rank      | 5.000     | 15.000    | 12.000    | 9.000     | 3.000            | 11.000    | 4.000            | 6.000            | 17.000    |
| FIR7    | Mean      | 2.366e-03 | 2.827e-03 | 2.779e-03 | 3.132e-03 | <b>1.632e-03</b> | 1.791e-03 | 1.771e-03        | 2.211e-03        | 8.587e-03 |
|         | Std       | 1.608e-03 | 1.146e-03 | 2.422e-03 | 2.131e-03 | 8.330e-04        | 9.903e-04 | 8.822e-04        | 1.467e-03        | 6.302e-03 |
|         | Rank      | 10.000    | 15.000    | 13.000    | 16.000    | 1.000            | 4.000     | 2.000            | 6.000            | 17.000    |
| FIR8    | Mean      | 6.060e-03 | 5.913e-03 | 7.095e-03 | 6.176e-03 | <b>3.602e-03</b> | 4.604e-03 | 4.450e-03        | 4.475e-03        | 9.224e-03 |
|         | Std       | 2.649e-03 | 3.262e-03 | 3.797e-03 | 2.782e-03 | 1.131e-03        | 1.703e-03 | 2.185e-03        | 2.038e-03        | 3.131e-03 |
|         | Rank      | 14.000    | 13.000    | 16.000    | 15.000    | 1.000            | 7.000     | 3.000            | 5.000            | 17.000    |
| AvgRank | Rank      | 12.125    | 15.250    | 13.250    | 14.125    | <b>4.750</b>     | 7.625     | 6.375            | 6.375            | 17.000    |



**Fig. 4.** Mean convergence curves on the eight FIR design cases for the internal-validation configurations and the MPHBS baseline. The horizontal axis reports function evaluations, and the vertical axis reports fitness value in logarithmic scale. Each curve corresponds to the average of 30 paired runs.

comparisons in Table 7 are significant against BR-C1 ( $p_{\text{Holm}} = 0.0043$ ) and R-best ( $p_{\text{Holm}} = 0.0191$ ). The instance-level paired tests provide complementary evidence: after correcting the complete CEC2022 ablation family, eight functions significantly favor B-C1 against BR-C1 and nine favor it against R-best, whereas one function significantly favors each RANDOM control. This result indicates that, in CEC2022, the learned mediator contributes beyond the matched structured randomized exchange mechanism (see Table 6).

The comparison against MPHBS is more nuanced. B-C1 and MPHBS each lead six functions, and the Friedman–Holm post hoc comparison is not significant ( $p_{\text{Holm}} = 0.8900$ ). Although four individual functions significantly favor B-C1 after the domain-wide correction, this evidence is not sufficiently consistent across the complete benchmark to establish a global rank difference. The HBA–MPA hybrid backbone is therefore already highly competitive in CEC2022, while the clearest incremental contribution of SARSA appears against the matched RANDOM alternatives.

The comparisons against the parent optimizers require a similarly differentiated reading. B-C1 wins all 12 functions against HBA and the corrected post hoc comparison is significant, with nine individual functions also significantly favoring the anchor. Against MPA, B-C1 wins eight functions and four instance-level tests favor the anchor, but the global Friedman–Holm comparison is not significant. Thus, the complete architecture clearly improves upon HBA, while MPA remains part of the leading statistical group on this heterogeneous benchmark.

Taken together, the CEC2022 ablation provides strong internal evidence for the learned mediator relative to its structured randomized controls. The omnibus Friedman test is significant and Kendall’s  $W = 0.530$  indicates a large separation in method ordering across functions. B-C1 achieves the best average rank and significantly surpasses BR-C1, R-best, and HBA, while its differences from MPHBS and MPA are not significant at the function-block level. The contribution of SARSA should therefore be interpreted as a meaningful improvement over matched randomized information exchange rather than as uniform superiority over every strong backbone or parent reference.

#### 5.4.4. FIR ablation study

The FIR ablation study was conducted using the final A-C1 configuration of MPHBS, reported as AS-C1 in the ablation tables, which

was selected in the previous sensitivity analysis as the best-ranked SARSA configuration for this domain. The purpose of this stage is to separate the contribution of the selected SARSA-based mediator from the contribution of the remaining architectural elements. To this end, the ablation compares the selected anchor against its matched RANDOM mirror, the best RANDOM configuration identified in the same domain, the MPHBS baseline, and the two parent optimizers HBA and MPA.

A first important result is that A-C1 provides the lowest global mean fitness in the FIR ablation set. The detailed case-wise results are reported in Table 8. It shares the best average rank with R-best and achieves the best mean fitness in four of the eight FIR cases, whereas R-best leads three cases and the matched RANDOM mirror leads one. Therefore, the selected SARSA-based configuration provides the best global mean and a strong case-wise balance, although it is not uniformly superior to the RANDOM controls.

However, the comparison against the RANDOM counterparts should be interpreted carefully. A-C1 obtains the lowest global mean fitness and wins six of eight FIR cases against AR-C1, but the Friedman–Holm post hoc comparison is not significant. Against R-best, both methods win four cases and share the same average rank of 1.750. Moreover, no instance-level Wilcoxon comparison against either RANDOM control remains significant after the domain-wide Holm correction. Consequently, the FIR results do not support a claim that SARSA is globally superior to the matched structured randomized exchange mechanism (see Table 9).

The detailed run-level W/T/L results in Table 10 are consistent with this block-level conclusion. The matched RANDOM comparison remains relatively balanced within several cases, while R-best alternates with the anchor across the benchmark. Therefore, the selected SARSA-based configuration does not dominate the RANDOM controls in a statistically decisive way, although it retains the lowest global mean fitness and a tied best average rank.

The comparison against MPHBS, HBA, and MPA is much clearer. A-C1 wins all eight FIR cases against each reference, every case-level Wilcoxon comparison remains significant after domain-wide Holm correction, and all three Friedman–Holm post hoc comparisons reject equality. Since MPHBS represents the hybrid backbone without the mediator phase, this consistent gap indicates that, in FIR, the main

**Table 5**

CEC2022 ablation results for the selected anchor, its matched RANDOM mirror, the best RANDOM configuration, MPHBS, HBA, and MPA. For each function, the error mean, standard deviation, and global rank are reported in separate rows. Global ranks are computed from the full ablation set using mean values. Lower values are better. The minimum mean within each displayed row block is highlighted in bold.

| Function | Statistic | BS-C1            | BR-C1            | R-best    | MPHB             | HBA       | MPA              |
|----------|-----------|------------------|------------------|-----------|------------------|-----------|------------------|
| F1       | Mean      | 3.615e-07        | <b>1.005e-09</b> | 4.916e-09 | 3.869e-04        | 3.548e-05 | 9.622e-05        |
|          | Std       | 2.687e-07        | 4.202e-09        | 1.529e-08 | 3.925e-04        | 1.832e-04 | 1.756e-04        |
|          | Rank      | 3.000            | 1.000            | 2.000     | 6.000            | 4.000     | 5.000            |
| F2       | Mean      | <b>2.752e+01</b> | 3.612e+01        | 4.007e+01 | 3.043e+01        | 3.761e+01 | 3.329e+01        |
|          | Std       | 2.399e+01        | 2.062e+01        | 1.830e+01 | 2.219e+01        | 2.497e+01 | 2.349e+01        |
|          | Rank      | 1.000            | 4.000            | 6.000     | 2.000            | 5.000     | 3.000            |
| F3       | Mean      | <b>1.803e-05</b> | 1.068e+00        | 1.002e+00 | 4.295e-02        | 1.385e+00 | 1.212e-02        |
|          | Std       | 2.712e-05        | 9.016e-01        | 1.043e+00 | 8.786e-02        | 2.328e+00 | 1.813e-02        |
|          | Rank      | 1.000            | 5.000            | 4.000     | 3.000            | 6.000     | 2.000            |
| F4       | Mean      | 3.735e+01        | 4.319e+01        | 4.645e+01 | <b>3.598e+01</b> | 5.641e+01 | 3.842e+01        |
|          | Std       | 1.087e+01        | 1.113e+01        | 1.293e+01 | 8.547e+00        | 1.312e+01 | 9.445e+00        |
|          | Rank      | 2.000            | 4.000            | 5.000     | 1.000            | 6.000     | 3.000            |
| F5       | Mean      | 3.031e+01        | 3.445e+02        | 3.181e+02 | <b>2.096e+01</b> | 2.291e+02 | 3.089e+01        |
|          | Std       | 3.510e+01        | 1.925e+02        | 1.723e+02 | 2.341e+01        | 2.982e+02 | 4.742e+01        |
|          | Rank      | 2.000            | 6.000            | 5.000     | 1.000            | 4.000     | 3.000            |
| F6       | Mean      | 2.264e+01        | 5.705e+01        | 5.576e+01 | 1.635e+01        | 7.518e+03 | <b>1.588e+01</b> |
|          | Std       | 1.691e+01        | 4.305e+01        | 3.510e+01 | 1.075e+01        | 7.839e+03 | 9.260e+00        |
|          | Rank      | 3.000            | 5.000            | 4.000     | 2.000            | 6.000     | 1.000            |
| F7       | Mean      | 2.945e+01        | 7.219e+01        | 6.164e+01 | 2.881e+01        | 5.364e+01 | <b>2.450e+01</b> |
|          | Std       | 1.045e+01        | 4.265e+01        | 3.215e+01 | 5.945e+00        | 2.024e+01 | 4.449e+00        |
|          | Rank      | 3.000            | 6.000            | 5.000     | 2.000            | 4.000     | 1.000            |
| F8       | Mean      | 2.056e+01        | 3.485e+01        | 2.967e+01 | 2.114e+01        | 4.264e+01 | <b>2.049e+01</b> |
|          | Std       | 2.297e+00        | 3.046e+01        | 2.191e+01 | 5.232e-01        | 4.112e+01 | 3.676e+00        |
|          | Rank      | 2.000            | 5.000            | 4.000     | 3.000            | 6.000     | 1.000            |
| F9       | Mean      | <b>1.808e+02</b> | 1.808e+02        | 1.808e+02 | 1.808e+02        | 1.808e+02 | 1.808e+02        |
|          | Std       | 1.049e-08        | 9.542e-07        | 1.851e-07 | 3.823e-06        | 5.517e-02 | 1.928e-06        |
|          | Rank      | 1.000            | 3.000            | 2.000     | 5.000            | 6.000     | 4.000            |
| F10      | Mean      | <b>5.154e+01</b> | 1.613e+02        | 1.365e+02 | 1.005e+02        | 1.293e+03 | 1.088e+02        |
|          | Std       | 5.894e+01        | 9.872e+01        | 8.971e+01 | 9.948e-02        | 1.411e+03 | 3.140e+01        |
|          | Rank      | 1.000            | 5.000            | 4.000     | 2.000            | 6.000     | 3.000            |
| F11      | Mean      | 3.000e+02        | 3.274e+02        | 3.267e+02 | <b>2.800e+02</b> | 3.513e+02 | 3.167e+02        |
|          | Std       | 6.433e+01        | 4.471e+01        | 4.498e+01 | 1.186e+02        | 2.082e+02 | 1.487e+02        |
|          | Rank      | 2.000            | 5.000            | 4.000     | 1.000            | 6.000     | 3.000            |
| F12      | Mean      | 2.410e+02        | 2.554e+02        | 2.548e+02 | <b>2.391e+02</b> | 3.954e+02 | 2.401e+02        |
|          | Std       | 7.529e+00        | 1.847e+01        | 1.614e+01 | 4.469e+00        | 1.158e+02 | 4.266e+00        |
|          | Rank      | 3.000            | 5.000            | 4.000     | 1.000            | 6.000     | 2.000            |
| AvgRank  | Rank      | <b>2.000</b>     | 4.500            | 4.083     | 2.417            | 5.417     | 2.583            |

**Table 6**

Function-wise W/T/L results for the CEC2022 ablation study. Each cell reports the wins/ties/losses of the column method against the selected anchor BS-C1 over 30 paired runs.

| Function | BR-C1   | R-best  | MPHB    | HBA     | MPA     |
|----------|---------|---------|---------|---------|---------|
| F1       | 30/0/0  | 30/0/0  | 0/0/30  | 22/0/8  | 0/0/30  |
| F2       | 15/0/15 | 17/0/13 | 11/0/19 | 13/0/17 | 11/0/19 |
| F3       | 0/0/30  | 0/0/30  | 0/0/30  | 0/0/30  | 0/0/30  |
| F4       | 10/0/20 | 6/0/24  | 17/0/13 | 2/0/28  | 12/0/18 |
| F5       | 0/0/30  | 0/0/30  | 20/0/10 | 6/0/24  | 18/0/12 |
| F6       | 8/0/22  | 6/0/24  | 20/0/10 | 0/0/30  | 19/0/11 |
| F7       | 2/0/28  | 3/0/27  | 13/0/17 | 3/0/27  | 20/0/10 |
| F8       | 0/0/30  | 0/0/30  | 13/0/17 | 1/0/29  | 13/0/17 |
| F9       | 3/1/26  | 6/1/23  | 4/0/26  | 5/0/25  | 0/0/30  |
| F10      | 3/0/27  | 5/0/25  | 3/0/27  | 1/0/29  | 3/0/27  |
| F11      | 17/0/13 | 19/0/11 | 14/0/16 | 26/0/4  | 4/0/26  |
| F12      | 3/0/27  | 3/0/27  | 17/0/13 | 0/0/30  | 15/0/15 |

performance gain is strongly associated with the presence of the explicit information-exchange architecture rather than with simple parallel hybridization alone.

Taken together, the FIR ablation leads to a nuanced but positive conclusion. The omnibus effect is very large ( $W = 0.848$ ), A-C1 provides the lowest global mean fitness, and it clearly improves upon MPHBS, HBA, and MPA. At the same time, its differences from AR-C1 and

R-best are not statistically conclusive. The strongest FIR conclusion is therefore that the information-exchange architecture is beneficial relative to the backbone and parent optimizers, while the incremental contribution of SARSA over matched randomized exchange remains domain- and landscape-dependent.

### 5.5. Comparative evaluation against recent metaheuristics

The previous sensitivity and ablation analyses were used to select one final MPHBS configuration per benchmark domain. For CEC2022, the final configuration of MPHBS is denoted B-C1 and combines the BEFORE FAD placement with  $K = 7$ ,  $N_i = 3$ ,  $w_{best} = 0.050$ ,  $\gamma = 0.30$ , and  $\rho_{sub} = 0.25$ . For FIR, the final configuration of MPHBS is denoted A-C1 and combines the AFTER FAD placement with  $K = 3$ ,  $N_i = 9$ ,  $w_{best} = 0.050$ ,  $\gamma = 0.30$ , and  $\rho_{sub} = 0.25$ . Both configurations instantiate the same MPHBS architecture under domain-specific validated settings; they should therefore be interpreted as benchmark-dependent settings of one algorithm rather than as separate methods.

The external comparison includes three internal references and six external competitors. The internal references are MPHBS, which represents the HBA-MPA hybrid backbone without the SARSA-guided reinforcement-learning mediator, and the two parent optimizers HBA and MPA. These methods allow each final MPHBS configuration to be evaluated against the components from which it is constructed.

**Table 7**

CEC2022 ablation analysis using functions as independent blocks. W/T/L is reported from the anchor perspective using function-wise means. Friedman  $p = 6.480 \times 10^{-6}$  and Kendall's  $W = 0.530$ . Post hoc  $p$ -values are Holm-adjusted across competitors. Sig. inst. A/C reports the numbers of function-level paired Wilcoxon tests that remain significant after domain-wide Holm correction in favor of the anchor or competitor. Lower AvgRank is better.

| Method | Role           | W/T/L  | Anchor rank | Comp. rank | Holm $p$  | Reject | Sig. inst. A/C |
|--------|----------------|--------|-------------|------------|-----------|--------|----------------|
| BR-C1  | MATCHED_RANDOM | 11/0/1 | 2.000       | 4.500      | 4.252e-03 | 1      | 8/1            |
| R-best | BEST_RANDOM    | 11/0/1 | 2.000       | 4.083      | 1.913e-02 | 1      | 9/1            |
| MPHB   | BASELINE       | 6/0/6  | 2.000       | 2.417      | 8.900e-01 | 0      | 4/0            |
| HBA    | PARENT         | 12/0/0 | 2.000       | 5.417      | 3.848e-05 | 1      | 9/0            |
| MPA    | PARENT         | 8/0/4  | 2.000       | 2.583      | 8.900e-01 | 0      | 4/0            |

**Table 8**

FIR ablation results for the selected anchor, its matched RANDOM mirror, the best RANDOM configuration, MPHB, HBA, and MPA. For each case, the fitness mean, standard deviation, and global rank are reported in separate rows. Global ranks are computed from the full ablation set using mean values. Lower values are better. The minimum mean within each displayed row block is highlighted in bold.

| Case    | Statistic | AS-C1            | AR-C1            | R-best           | MPHB      | HBA       | MPA       |
|---------|-----------|------------------|------------------|------------------|-----------|-----------|-----------|
| FIR1    | Mean      | 5.739e-03        | 5.433e-03        | <b>5.155e-03</b> | 1.356e-02 | 1.782e-02 | 1.018e-02 |
|         | Std       | 1.704e-03        | 8.048e-04        | 6.102e-04        | 7.689e-03 | 2.466e-02 | 4.412e-03 |
|         | Rank      | 3.000            | 2.000            | 1.000            | 5.000     | 6.000     | 4.000     |
| FIR2    | Mean      | 2.147e-02        | <b>2.068e-02</b> | 2.101e-02        | 3.448e-02 | 4.034e-02 | 2.952e-02 |
|         | Std       | 2.619e-03        | 2.526e-03        | 2.175e-03        | 1.075e-02 | 3.096e-02 | 5.920e-03 |
|         | Rank      | 3.000            | 1.000            | 2.000            | 5.000     | 6.000     | 4.000     |
| FIR3    | Mean      | <b>5.157e-03</b> | 5.303e-03        | 5.339e-03        | 1.333e-02 | 1.777e-02 | 1.186e-02 |
|         | Std       | 6.868e-04        | 9.921e-04        | 1.049e-03        | 5.430e-03 | 1.709e-02 | 6.811e-03 |
|         | Rank      | 1.000            | 2.000            | 3.000            | 5.000     | 6.000     | 4.000     |
| FIR4    | Mean      | 2.149e-02        | 2.180e-02        | <b>2.131e-02</b> | 3.379e-02 | 4.389e-02 | 2.901e-02 |
|         | Std       | 2.036e-03        | 2.981e-03        | 2.216e-03        | 1.113e-02 | 3.166e-02 | 7.167e-03 |
|         | Rank      | 2.000            | 3.000            | 1.000            | 5.000     | 6.000     | 4.000     |
| FIR5    | Mean      | <b>1.760e-03</b> | 2.028e-03        | 1.822e-03        | 6.874e-03 | 1.233e-02 | 5.217e-03 |
|         | Std       | 6.678e-04        | 7.823e-04        | 7.081e-04        | 3.863e-03 | 1.637e-02 | 3.509e-03 |
|         | Rank      | 1.000            | 3.000            | 2.000            | 5.000     | 6.000     | 4.000     |
| FIR6    | Mean      | 6.668e-03        | 6.979e-03        | <b>6.663e-03</b> | 1.235e-02 | 1.157e-02 | 9.987e-03 |
|         | Std       | 1.672e-03        | 1.534e-03        | 1.651e-03        | 5.179e-03 | 4.373e-03 | 3.866e-03 |
|         | Rank      | 2.000            | 3.000            | 1.000            | 6.000     | 5.000     | 4.000     |
| FIR7    | Mean      | <b>1.632e-03</b> | 2.271e-03        | 1.775e-03        | 8.587e-03 | 7.697e-03 | 8.943e-03 |
|         | Std       | 8.330e-04        | 1.484e-03        | 1.237e-03        | 6.302e-03 | 8.204e-03 | 8.669e-03 |
|         | Rank      | 1.000            | 3.000            | 2.000            | 5.000     | 4.000     | 6.000     |
| FIR8    | Mean      | <b>3.602e-03</b> | 4.819e-03        | 4.473e-03        | 9.224e-03 | 2.379e-02 | 7.485e-03 |
|         | Std       | 1.131e-03        | 2.598e-03        | 2.343e-03        | 3.131e-03 | 2.381e-02 | 2.647e-03 |
|         | Rank      | 1.000            | 3.000            | 2.000            | 5.000     | 6.000     | 4.000     |
| AvgRank | Rank      | <b>1.750</b>     | 2.500            | 1.750            | 5.125     | 5.625     | 4.250     |

**Table 9**

FIR ablation analysis using cases as independent blocks. W/T/L is reported from the anchor perspective using case-wise means. Friedman  $p = 2.460 \times 10^{-6}$  and Kendall's  $W = 0.848$ . Post hoc  $p$ -values are Holm-adjusted across competitors. Sig. inst. A/C reports the numbers of case-level paired Wilcoxon tests that remain significant after domain-wide Holm correction in favor of the anchor or competitor. Lower AvgRank is better.

| Method | Role           | W/T/L | Anchor rank | Comp. rank | Holm $p$  | Reject | Sig. inst. A/C |
|--------|----------------|-------|-------------|------------|-----------|--------|----------------|
| AR-C1  | MATCHED_RANDOM | 6/0/2 | 1.750       | 2.500      | 8.454e-01 | 0      | 0/0            |
| R-best | BEST_RANDOM    | 4/0/4 | 1.750       | 1.750      | 1.000e+00 | 0      | 0/0            |
| MPHB   | BASELINE       | 8/0/0 | 1.750       | 5.125      | 1.234e-03 | 1      | 8/0            |
| HBA    | PARENT         | 8/0/0 | 1.750       | 5.625      | 1.717e-04 | 1      | 8/0            |
| MPA    | PARENT         | 8/0/0 | 1.750       | 4.250      | 2.258e-02 | 1      | 8/0            |

The external competitors are Hybrid Equilibrium-Differential Evolution (EODE) [23,24], Reinforcement-Learning Teaching-Learning-Based Optimization (RLTLBO) [25], Quasi-opposition and Q-learning Marine Predators Algorithm (QLMPA) [26], Grey Wolf Optimizer (GWO) [27], fitness-distance-balance teaching-learning artificial bee colony (FDB-TLABC) [28], and Fourier Transform Optimizer (FTO) [29]. EODE is included as a recent hybrid DE-based external comparator, representing a strong and non-trivial DE-related search line against MPHBs. RLTLBO and QLMPA are included as additional learning-based meta-heuristic comparators, which help separate the proposed SARSA-guided component-exchange mechanism from the more general idea of adding learning to a population-based optimizer. GWO is included as a classical

and widely recognizable baseline, FDB-TLABC as a recent population-based alternative, and FTO as a recent physics-inspired alternative with a distinct search mechanism.

This comparison is stricter than the sensitivity and ablation stages because the selected MPHBs configurations are no longer being tuned or interpreted internally. Instead, B-C1 and A-C1 are fixed before the external evaluation and are compared against independent baselines under the same population size, number of paired runs, and function-evaluation budgets used in each benchmark domain. Therefore, the external stage assesses whether the configurations selected by the internal validation remain competitive when confronted with methods outside the immediate MPHBs design space.

**Table 10**

Case-wise W/T/L results for the FIR ablation study. Each cell reports the wins/ties/losses of the column method against the selected anchor AS-C1 over 30 paired runs.

| Case | AR-C1   | R-best  | MPHB   | HBA    | MPA    |
|------|---------|---------|--------|--------|--------|
| FIR1 | 12/0/18 | 17/0/13 | 2/0/28 | 4/0/26 | 3/0/27 |
| FIR2 | 17/0/13 | 17/0/13 | 1/0/29 | 3/0/27 | 2/0/28 |
| FIR3 | 15/0/15 | 16/0/14 | 0/0/30 | 1/0/29 | 2/0/28 |
| FIR4 | 16/0/14 | 15/0/15 | 0/0/30 | 1/0/29 | 4/0/26 |
| FIR5 | 12/0/18 | 14/0/16 | 0/0/30 | 3/0/27 | 1/0/29 |
| FIR6 | 12/0/18 | 17/0/13 | 0/0/30 | 4/0/26 | 8/0/22 |
| FIR7 | 11/0/19 | 15/0/15 | 0/0/30 | 5/0/25 | 0/0/30 |
| FIR8 | 11/0/19 | 13/0/17 | 1/0/29 | 1/0/29 | 1/0/29 |

The interpretation follows three complementary criteria. First, mean and standard deviation values assess final solution quality and run-to-run stability. Second, average ranks, instance-level W/T/L summaries, the Friedman omnibus test, and Kendall's  $W$  assess consistency and global effect magnitude across functions or cases. Third, Wilcoxon signed-rank tests evaluate the 30 seed-matched outcomes within each instance, while Friedman–Holm post hoc comparisons evaluate anchor differences at the multi-problem level. This separation prevents repeated runs from being treated as independent benchmark problems.

#### 5.5.1. CEC2022 external comparison

Table 11 reports the external comparison on the 12 CEC2022 functions. The results show that the final CEC2022 configuration of MPHBS, denoted B-C1, is the most balanced method in the enlarged comparison set according to the mean-based average rank. B-C1 obtains an average rank of 2.750, followed by MPHBS with 3.000 and MPA with 3.250; EODE, FDB-TLABC, and QQLMPA obtain 4.000, 4.417, and 5.167, respectively. This places the selected configuration at the front of the leading mean-rank group while retaining the heterogeneous function-wise behavior expected on CEC2022.

At the function level, the results show that no single method dominates the complete CEC2022 benchmark. EODE achieves the best mean error in F1 and F5, FDB-TLABC obtains the best result in F4, and MPA provides the best means in F6–F8. MPHBS leads F2, F11, and F12, while RLTLBO records the lowest mean in F9. The final CEC2022 configuration of MPHBS obtains the best means in F3 and F10 and remains close to the strongest method in several additional functions. This distribution of wins indicates that CEC2022 produces a heterogeneous scenario in which different search mechanisms are favored by different landscape properties.

The comparison against the internal references provides a first layer of interpretation. B-C1 has an average rank close to MPHBS and MPA, while HBA is clearly weaker. At the function-block level, B-C1 and MPHBS each win six functions, and B-C1 wins eight against MPA; neither post hoc difference is significant. In contrast, B-C1 wins all 12 functions against HBA and the corrected comparison is significant. This suggests that the final CEC2022 configuration preserves the strength of the hybrid backbone and remains comparable to MPA while avoiding the weak global behavior of the isolated HBA parent.

The behavior of EODE deserves a careful interpretation. B-C1 wins eight functions, ties one, and loses three according to the function-wise means, while EODE remains highly competitive within several individual run distributions. The Friedman–Holm post hoc comparison is not significant ( $p_{\text{Holm}} = 0.9356$ ). At the same time, EODE has a substantially larger global mean error and a weaker mean-based average rank because of unstable behavior on difficult functions such as F6 and F10. Therefore, EODE and B-C1 belong to the leading statistical group, with B-C1 providing the stronger aggregate error and mean-rank balance.

The remaining external competitors exhibit more limited behavior. FDB-TLABC and QQLMPA lose eight and ten functions, respectively,

but their Friedman–Holm post hoc comparisons do not reach significance after accounting for the complete method set. In contrast, B-C1 significantly outperforms RLTLBO, GWO, and FTO at the function-block level, with corresponding W/T/L counts of 11/0/1, 12/0/0, and 12/0/0. These results preserve the conclusion that MPHBS is competitive against diverse external methods without implying uniform superiority over every comparator.

The convergence curves in Fig. 5 provide a complementary view of this behavior by showing how the methods progress under the same function-evaluation budget. These curves are used to support the interpretation of final accuracy with the temporal behavior of the search process, especially in functions where strong final performance is associated with faster or more stable convergence.

The detailed run-level W/T/L results in Table 12 reinforce the heterogeneous character of CEC2022. EODE wins many paired runs on F1 and F5, whereas B-C1 has pronounced advantages on F6, F8, and F10. MPHBS and MPA also alternate with the anchor across functions. These local patterns explain why strong instance-level differences can coexist with non-significant global post hoc comparisons when functions, rather than repeated runs, are treated as the independent blocks.

The block-level analysis in Table 13 confirms this nuanced outcome. The omnibus difference is significant with a large concordance effect ( $W = 0.626$ ). B-C1 is statistically indistinguishable from MPHBS, MPA, EODE, QQLMPA, and FDB-TLABC after Holm correction, while it significantly outperforms HBA, RLTLBO, GWO, and FTO. Consequently, the strongest CEC2022 conclusion is membership in the leading statistical group together with the best mean-based average rank, rather than universal dominance.

#### 5.5.2. FIR external comparison

Table 14 reports the external comparison on the eight FIR filter design cases. In this benchmark, the final FIR configuration of MPHBS, denoted A-C1, is the best-performing method in every case and obtains the best overall average rank of 1.000. EODE follows with 2.625 and MPA with 3.125, while QQLMPA and RLTLBO remain behind the leading group with average ranks of 6.750 and 8.625, respectively. This result indicates that the final FIR configuration is stronger than the internal baseline, the parent optimizers, the recent hybrid DE-based comparator, and the added learning-based metaheuristic competitors under the adopted FIR design protocol.

The FIR results differ from the CEC2022 results in an important way. Whereas CEC2022 produces a balanced scenario with several competitive methods, the structured FIR benchmark shows a clear and consistent advantage for A-C1 under the adopted FIR design protocol. The final FIR configuration of MPHBS obtains the best mean fitness in all eight FIR cases, including low-pass, high-pass, band-pass, and band-stop designs. This indicates that the selected AFTER configuration is not specialized to a single filter type; instead, it maintains strong behavior across different spectral structures and transition-band difficulties.

The comparison against internal references is also more decisive than in CEC2022. MPHBS, HBA, and MPA are all clearly worse than A-C1 in terms of mean fitness and average rank. Among them, MPA is the strongest internal reference, with an average rank of 3.125, but it remains substantially behind A-C1. This result is consistent with the ablation analysis: for FIR, the full information-exchange architecture provides a stronger gain over the hybrid backbone and over the isolated parent optimizers than in the CEC2022 setting.

Among the external competitors, EODE is the closest method to the selected MPHBS configuration, with the second-best average rank. This indicates that a hybrid DE-based search strategy is also well suited to the FIR design landscape. However, EODE does not surpass A-C1 in any FIR case according to the mean values. QQLMPA and RLTLBO remain informative learning-based comparators: despite incorporating learning mechanisms, they do not approach the performance of the SARSA-guided HBA–MPA information-exchange architecture.

**Table 11**

CEC2022 external comparative results for the final proposed method, internal reference baselines, and selected external competitors. For each function, the error mean, standard deviation, and global rank are reported in separate rows. Global ranks are computed from the full external comparison set using mean values. Lower values are better. The minimum mean within each displayed row block is highlighted in bold.

| Function | Statistic | B-C1             | MPHB             | HBA       | MPA              | EODE             | RLTLBO           | QQLMPA    | GWO       | FDB-TLABC        | FTO       |
|----------|-----------|------------------|------------------|-----------|------------------|------------------|------------------|-----------|-----------|------------------|-----------|
| F1       | Mean      | 3.615e-07        | 3.869e-04        | 3.548e-05 | 9.622e-05        | <b>3.888e-11</b> | 4.906e-01        | 8.403e+00 | 9.288e+03 | 8.281e-06        | 8.341e+03 |
|          | Std       | 2.687e-07        | 3.925e-04        | 1.832e-04 | 1.756e-04        | 1.284e-10        | 1.251e+00        | 1.981e+01 | 3.660e+03 | 4.528e-05        | 3.383e+03 |
|          | Rank      | 2.000            | 6.000            | 4.000     | 5.000            | 1.000            | 7.000            | 8.000     | 10.000    | 3.000            | 9.000     |
| F2       | Mean      | <b>2.752e+01</b> | 3.043e+01        | 3.761e+01 | 3.329e+01        | 4.091e+01        | 6.688e+01        | 4.532e+01 | 8.851e+01 | 4.820e+01        | 1.160e+02 |
|          | Std       | 2.399e+01        | 2.219e+01        | 2.497e+01 | 2.349e+01        | 1.824e+01        | 2.406e+01        | 1.917e+01 | 3.566e+01 | 1.542e+01        | 3.527e+01 |
|          | Rank      | 1.000            | 2.000            | 4.000     | 3.000            | 5.000            | 8.000            | 6.000     | 9.000     | 7.000            | 10.000    |
| F3       | Mean      | <b>1.803e-05</b> | 4.295e-02        | 1.385e+00 | 1.212e-02        | 7.473e-03        | 7.586e+00        | 7.082e-01 | 4.294e+00 | 9.529e-01        | 3.002e+01 |
|          | Std       | 2.712e-05        | 8.786e-02        | 2.328e+00 | 1.813e-02        | 3.456e-02        | 6.128e+00        | 3.437e-01 | 3.072e+00 | 1.205e+00        | 7.580e+00 |
|          | Rank      | 1.000            | 4.000            | 7.000     | 3.000            | 2.000            | 9.000            | 5.000     | 8.000     | 6.000            | 10.000    |
| F4       | Mean      | 3.735e+01        | 3.598e+01        | 5.641e+01 | 3.842e+01        | 2.704e+01        | 7.643e+01        | 3.461e+01 | 4.748e+01 | <b>2.404e+01</b> | 5.633e+01 |
|          | Std       | 1.087e+01        | 8.547e+00        | 1.312e+01 | 9.445e+00        | 1.099e+01        | 1.727e+01        | 8.962e+00 | 1.639e+01 | 6.178e+00        | 3.227e+01 |
|          | Rank      | 5.000            | 4.000            | 9.000     | 6.000            | 2.000            | 10.000           | 3.000     | 7.000     | 1.000            | 8.000     |
| F5       | Mean      | 3.031e+01        | 2.096e+01        | 2.291e+02 | 3.089e+01        | <b>2.767e-01</b> | 1.938e+02        | 2.514e+01 | 1.920e+02 | 2.943e+01        | 3.359e+02 |
|          | Std       | 3.510e+01        | 2.341e+01        | 2.982e+02 | 4.742e+01        | 7.627e-01        | 2.095e+02        | 2.354e+01 | 1.978e+02 | 2.998e+01        | 2.108e+02 |
|          | Rank      | 5.000            | 2.000            | 9.000     | 6.000            | 1.000            | 8.000            | 3.000     | 7.000     | 4.000            | 10.000    |
| F6       | Mean      | 2.264e+01        | 1.635e+01        | 7.518e+03 | <b>1.588e+01</b> | 5.102e+03        | 2.351e+03        | 2.368e+01 | 9.266e+04 | 2.619e+03        | 4.073e+03 |
|          | Std       | 1.691e+01        | 1.075e+01        | 7.839e+03 | 9.260e+00        | 6.754e+03        | 3.373e+03        | 1.061e+01 | 1.833e+05 | 2.273e+03        | 5.867e+03 |
|          | Rank      | 3.000            | 2.000            | 9.000     | 1.000            | 8.000            | 5.000            | 4.000     | 10.000    | 6.000            | 7.000     |
| F7       | Mean      | 2.945e+01        | 2.881e+01        | 5.364e+01 | <b>2.450e+01</b> | 3.646e+01        | 6.083e+01        | 3.500e+01 | 8.335e+01 | 2.546e+01        | 1.063e+02 |
|          | Std       | 1.045e+01        | 5.945e+00        | 2.024e+01 | 4.449e+00        | 1.471e+01        | 1.796e+01        | 6.189e+00 | 4.696e+01 | 8.815e+00        | 4.327e+01 |
|          | Rank      | 4.000            | 3.000            | 7.000     | 1.000            | 6.000            | 8.000            | 5.000     | 9.000     | 2.000            | 10.000    |
| F8       | Mean      | 2.056e+01        | 2.114e+01        | 4.264e+01 | <b>2.049e+01</b> | 2.342e+01        | 3.879e+01        | 2.471e+01 | 5.595e+01 | 2.530e+01        | 3.142e+01 |
|          | Std       | 2.297e+00        | 5.232e-01        | 4.112e+01 | 3.676e+00        | 2.976e+00        | 3.048e+01        | 1.376e+00 | 5.021e+01 | 3.645e+00        | 6.598e+00 |
|          | Rank      | 2.000            | 3.000            | 9.000     | 1.000            | 4.000            | 8.000            | 5.000     | 10.000    | 6.000            | 7.000     |
| F9       | Mean      | 1.808e+02        | 1.808e+02        | 1.808e+02 | 1.808e+02        | 1.808e+02        | <b>1.808e+02</b> | 1.808e+02 | 2.078e+02 | 1.808e+02        | 1.901e+02 |
|          | Std       | 1.049e-08        | 3.823e-06        | 5.517e-02 | 1.928e-06        | 8.672e-14        | 5.251e-03        | 3.984e-05 | 1.513e+01 | 9.801e-13        | 1.011e+01 |
|          | Rank      | 4.000            | 6.000            | 8.000     | 5.000            | 2.000            | 1.000            | 10.000    | 10.000    | 3.000            | 9.000     |
| F10      | Mean      | <b>5.154e+01</b> | 1.005e+02        | 1.293e+03 | 1.088e+02        | 3.542e+02        | 2.092e+02        | 1.135e+02 | 8.048e+02 | 1.495e+02        | 8.104e+02 |
|          | Std       | 5.894e+01        | 9.948e-02        | 1.411e+03 | 3.140e+01        | 4.181e+02        | 3.516e+02        | 3.922e+01 | 6.425e+02 | 6.557e+01        | 5.242e+02 |
|          | Rank      | 1.000            | 2.000            | 10.000    | 3.000            | 7.000            | 6.000            | 4.000     | 8.000     | 5.000            | 9.000     |
| F11      | Mean      | 3.000e+02        | <b>2.800e+02</b> | 3.513e+02 | 3.167e+02        | 3.267e+02        | 3.377e+02        | 3.626e+02 | 7.819e+02 | 3.208e+02        | 6.370e+02 |
|          | Std       | 6.433e+01        | 1.186e+02        | 2.082e+02 | 1.487e+02        | 4.498e+01        | 1.125e+02        | 8.024e+01 | 2.522e+02 | 4.051e+01        | 1.756e+02 |
|          | Rank      | 2.000            | 1.000            | 7.000     | 3.000            | 5.000            | 6.000            | 10.000    | 10.000    | 4.000            | 9.000     |
| F12      | Mean      | 2.410e+02        | <b>2.391e+02</b> | 3.954e+02 | 2.401e+02        | 2.425e+02        | 2.696e+02        | 2.411e+02 | 2.688e+02 | 2.612e+02        | 2.687e+02 |
|          | Std       | 7.529e+00        | 4.469e+00        | 1.158e+02 | 4.266e+00        | 5.227e+00        | 3.122e+01        | 4.396e+00 | 2.159e+01 | 1.550e+01        | 1.591e+01 |
|          | Rank      | 3.000            | 1.000            | 10.000    | 2.000            | 5.000            | 9.000            | 4.000     | 8.000     | 6.000            | 7.000     |
| AvgRank  | Rank      | <b>2.750</b>     | 3.000            | 7.750     | 3.250            | 4.000            | 7.083            | 5.167     | 8.833     | 4.417            | 8.750     |

**Table 12**

Function-wise W/T/L results for the CEC2022 external comparison. Each cell reports the wins/ties/losses of the column method against the final CEC2022 configuration of MPHBS, B-C1, over 30 paired runs.

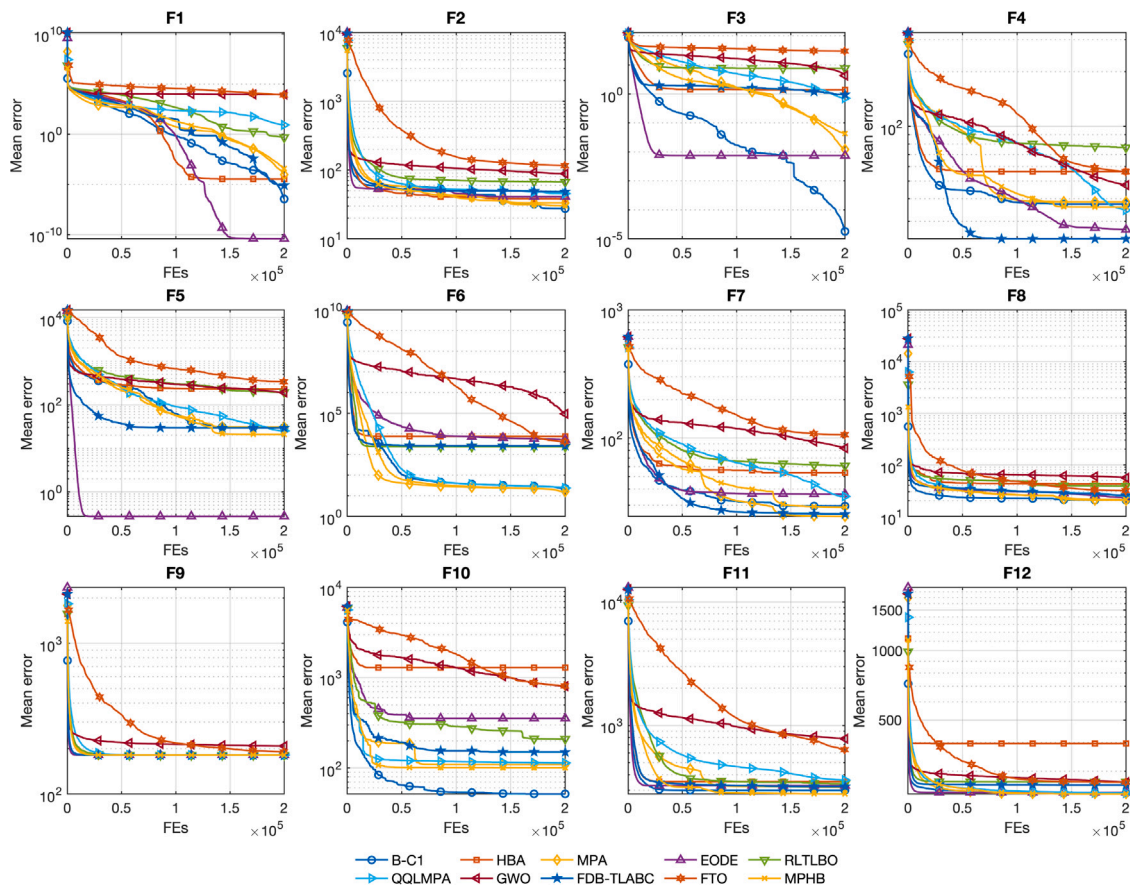
| Function | MPHB    | HBA     | MPA     | EODE    | RLTLBO  | QQLMPA  | GWO    | FDB-TLABC | FTO    |
|----------|---------|---------|---------|---------|---------|---------|--------|-----------|--------|
| F1       | 0/0/30  | 22/0/8  | 0/0/30  | 30/0/0  | 0/0/30  | 0/0/30  | 0/0/30 | 29/0/1    | 0/0/30 |
| F2       | 11/0/19 | 13/0/17 | 11/0/19 | 16/0/14 | 1/0/29  | 6/0/24  | 1/0/29 | 8/0/22    | 0/0/30 |
| F3       | 0/0/30  | 0/0/30  | 0/0/30  | 12/0/18 | 0/0/30  | 0/0/30  | 0/0/30 | 0/0/30    | 0/0/30 |
| F4       | 17/0/13 | 2/0/28  | 12/0/18 | 22/0/8  | 1/0/29  | 17/0/13 | 8/0/22 | 27/0/3    | 8/0/22 |
| F5       | 20/0/10 | 6/0/24  | 18/0/12 | 30/0/0  | 3/0/27  | 16/0/14 | 4/0/26 | 16/0/14   | 2/0/28 |
| F6       | 20/0/10 | 0/0/30  | 19/0/11 | 0/0/30  | 0/0/30  | 14/0/16 | 0/0/30 | 0/0/30    | 0/0/30 |
| F7       | 13/0/17 | 3/0/27  | 20/0/10 | 11/0/19 | 2/0/28  | 9/0/21  | 2/0/28 | 19/0/11   | 0/0/30 |
| F8       | 13/0/17 | 1/0/29  | 13/0/17 | 2/0/28  | 0/0/30  | 0/0/30  | 0/0/30 | 4/0/26    | 0/0/30 |
| F9       | 4/0/26  | 5/0/25  | 0/0/30  | 28/2/0  | 16/0/14 | 0/0/30  | 0/0/30 | 25/4/1    | 0/0/30 |
| F10      | 3/0/27  | 1/0/29  | 3/0/27  | 8/0/22  | 1/0/29  | 2/0/28  | 2/0/28 | 2/0/28    | 0/0/30 |
| F11      | 14/0/16 | 26/0/4  | 4/0/26  | 22/0/8  | 2/0/28  | 2/0/28  | 0/0/30 | 25/0/5    | 1/0/29 |
| F12      | 17/0/13 | 0/0/30  | 15/0/15 | 10/0/20 | 3/0/27  | 11/0/19 | 2/0/28 | 1/0/29    | 1/0/29 |

FDB-TLABC, GWO, and FTO are less competitive overall, especially in the harder or narrower designs where their final fitness values and ranks deteriorate more clearly. Therefore, the external FIR comparison provides stronger evidence in favor of MPHBS than the CEC2022 comparison.

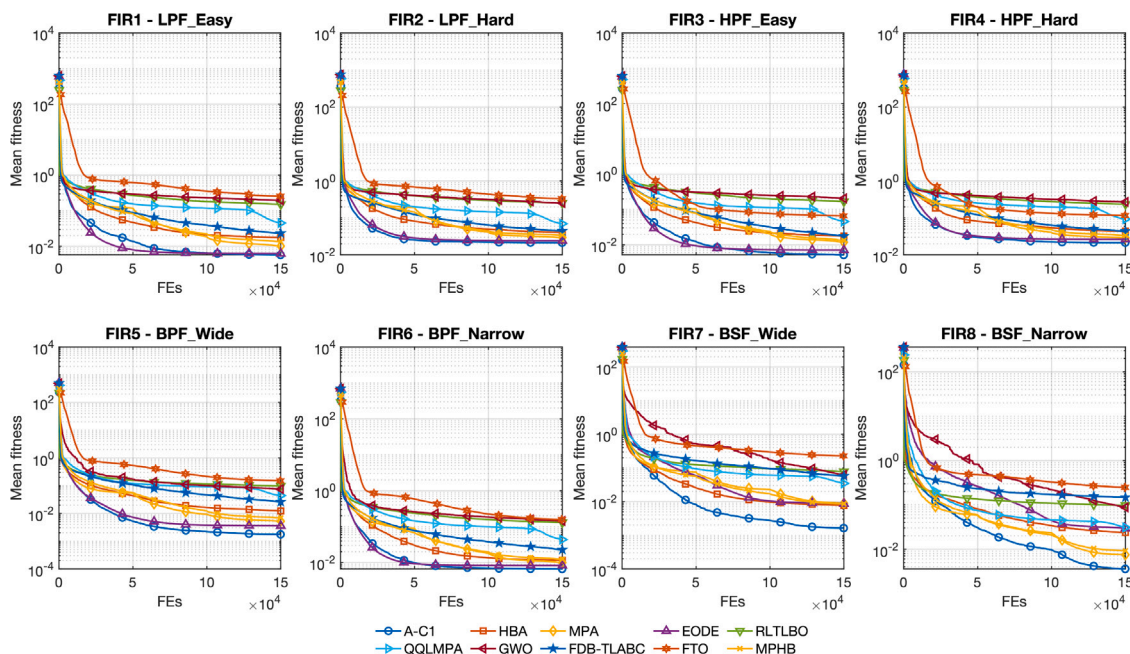
The convergence profiles in Fig. 6 further support this result. Across the FIR cases, the final FIR configuration of MPHBS reaches low fitness values consistently under the adopted evaluation budget, which agrees with the mean-rank and W/T/L evidence reported in the external

comparison tables. Although FDB-TLABC exhibits competitive convergence behavior in some panels, especially during intermediate stages of the search, the final interpretation is based on the complete set of final fitness values, average ranks, W/T/L counts, instance-level paired Wilcoxon tests, and the Friedman–Holm block analysis reported in the accompanying tables.

The case-wise W/T/L results in Table 15 show that A-C1 is consistently favored within the paired runs. EODE is the closest external competitor and records 49 wins against 191 losses, while QQLMPA records only two wins against 238 losses. At the case level, A-C1 obtains



**Fig. 5.** Mean convergence curves on the 12 CEC2022 benchmark functions for the final CEC2022 configuration of MPHBS, B-C1, internal reference baselines, and external competitors. The horizontal axis reports function evaluations, and the vertical axis reports residual error in logarithmic scale. Each curve corresponds to the average of 30 paired runs.



**Fig. 6.** Mean convergence curves on the eight FIR design cases for the final FIR configuration of MPHBS, A-C1, internal reference baselines, and external competitors. The horizontal axis reports function evaluations, and the vertical axis reports fitness value in logarithmic scale. Each curve corresponds to the average of 30 paired runs.

**Table 13**

CEC2022 external comparison using functions as independent blocks. W/T/L is reported from the anchor perspective using function-wise means. Friedman  $p = 4.454 \times 10^{-11}$  and Kendall's  $W = 0.626$ . Post hoc  $p$ -values are Holm-adjusted across competitors. Sig. inst. A/C reports the numbers of function-level paired Wilcoxon tests that remain significant after domain-wide Holm correction in favor of the anchor or competitor. Lower AvgRank is better.

| Method    | Role     | W/T/L  | Anchor rank | Comp. rank | Holm $p$  | Reject | Sig. inst. A/C |
|-----------|----------|--------|-------------|------------|-----------|--------|----------------|
| MPHB      | BASELINE | 6/0/6  | 2.750       | 3.000      | 1.000e+00 | 0      | 4/0            |
| HBA       | PARENT   | 12/0/0 | 2.750       | 7.750      | 3.660e-04 | 1      | 9/0            |
| MPA       | PARENT   | 8/0/4  | 2.750       | 3.250      | 1.000e+00 | 0      | 4/0            |
| EODE      | EXTERNAL | 8/1/3  | 2.750       | 4.000      | 9.356e-01 | 0      | 3/2            |
| RLTLBO    | EXTERNAL | 11/0/1 | 2.750       | 7.083      | 2.731e-03 | 1      | 11/0           |
| QQLMPA    | EXTERNAL | 10/0/2 | 2.750       | 5.167      | 2.528e-01 | 0      | 7/0            |
| GWO       | EXTERNAL | 12/0/0 | 2.750       | 8.833      | 7.723e-06 | 1      | 11/0           |
| FDB-TLABC | EXTERNAL | 8/1/3  | 2.750       | 4.417      | 7.101e-01 | 0      | 7/1            |
| FTO       | EXTERNAL | 12/0/0 | 2.750       | 8.750      | 9.668e-06 | 1      | 12/0           |

**Table 14**

FIR external comparative results for the final proposed method, internal reference baselines, and selected external competitors. For each case, the fitness Mean, standard deviation, and global rank are reported in separate rows. Global ranks are computed from the full external comparison set using mean values. Lower values are better. The minimum mean within each displayed row block is highlighted in bold.

| Case    | Statistic | A-C1             | MPHB      | HBA       | MPA       | EODE      | RLTLBO    | QQLMPA    | GWO       | FDB-TLABC | FTO       |
|---------|-----------|------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| FIR1    | Mean      | <b>5.739e-03</b> | 1.356e-02 | 1.782e-02 | 1.018e-02 | 6.268e-03 | 1.514e-01 | 4.518e-02 | 1.974e-01 | 2.334e-02 | 2.571e-01 |
|         | Std       | 1.704e-03        | 7.689e-03 | 2.466e-02 | 4.412e-03 | 2.571e-03 | 1.078e-01 | 3.304e-02 | 1.104e-01 | 2.289e-02 | 6.669e-02 |
|         | Rank      | 1.000            | 4.000     | 5.000     | 3.000     | 2.000     | 8.000     | 7.000     | 9.000     | 6.000     | 10.000    |
| FIR2    | Mean      | <b>2.147e-02</b> | 3.448e-02 | 4.034e-02 | 2.952e-02 | 2.435e-02 | 2.548e-01 | 6.966e-02 | 2.537e-01 | 4.467e-02 | 3.324e-01 |
|         | Std       | 2.619e-03        | 1.075e-02 | 3.096e-02 | 5.920e-03 | 5.161e-03 | 8.560e-02 | 5.410e-02 | 1.148e-01 | 2.597e-02 | 6.228e-02 |
|         | Rank      | 1.000            | 4.000     | 5.000     | 3.000     | 2.000     | 9.000     | 7.000     | 8.000     | 6.000     | 10.000    |
| FIR3    | Mean      | <b>5.157e-03</b> | 1.333e-02 | 1.777e-02 | 1.186e-02 | 7.092e-03 | 1.679e-01 | 4.551e-02 | 2.070e-01 | 1.761e-02 | 6.580e-02 |
|         | Std       | 6.868e-04        | 5.430e-03 | 1.709e-02 | 6.811e-03 | 3.047e-03 | 1.087e-01 | 4.184e-02 | 1.136e-01 | 1.272e-02 | 5.210e-02 |
|         | Rank      | 1.000            | 4.000     | 6.000     | 3.000     | 2.000     | 9.000     | 7.000     | 10.000    | 5.000     | 8.000     |
| FIR4    | Mean      | <b>2.149e-02</b> | 3.379e-02 | 4.389e-02 | 2.901e-02 | 2.646e-02 | 2.347e-01 | 9.215e-02 | 2.711e-01 | 4.400e-02 | 1.164e-01 |
|         | Std       | 2.036e-03        | 1.113e-02 | 3.166e-02 | 7.167e-03 | 3.614e-03 | 1.168e-01 | 5.160e-02 | 1.277e-01 | 2.289e-02 | 4.677e-02 |
|         | Rank      | 1.000            | 4.000     | 5.000     | 3.000     | 2.000     | 9.000     | 7.000     | 10.000    | 6.000     | 8.000     |
| FIR5    | Mean      | <b>1.760e-03</b> | 6.874e-03 | 1.233e-02 | 5.217e-03 | 3.581e-03 | 9.647e-02 | 4.389e-02 | 7.514e-02 | 2.595e-02 | 1.494e-01 |
|         | Std       | 6.678e-04        | 3.863e-03 | 1.637e-02 | 3.509e-03 | 2.069e-03 | 6.111e-02 | 4.294e-02 | 6.570e-02 | 1.997e-02 | 7.815e-02 |
|         | Rank      | 1.000            | 4.000     | 5.000     | 3.000     | 2.000     | 9.000     | 7.000     | 8.000     | 6.000     | 10.000    |
| FIR6    | Mean      | <b>6.668e-03</b> | 1.235e-02 | 1.157e-02 | 9.987e-03 | 8.263e-03 | 1.296e-01 | 4.380e-02 | 1.434e-01 | 2.271e-02 | 1.567e-01 |
|         | Std       | 1.672e-03        | 5.179e-03 | 4.373e-03 | 3.866e-03 | 1.571e-03 | 7.109e-02 | 2.715e-02 | 1.086e-01 | 2.220e-02 | 8.256e-02 |
|         | Rank      | 1.000            | 5.000     | 4.000     | 3.000     | 2.000     | 8.000     | 7.000     | 9.000     | 6.000     | 10.000    |
| FIR7    | Mean      | <b>1.632e-03</b> | 8.587e-03 | 7.697e-03 | 8.943e-03 | 8.685e-03 | 7.759e-02 | 3.469e-02 | 6.240e-02 | 5.714e-02 | 2.279e-01 |
|         | Std       | 8.330e-04        | 6.302e-03 | 8.204e-03 | 8.669e-03 | 5.870e-03 | 3.370e-02 | 2.108e-02 | 4.224e-02 | 2.938e-02 | 4.883e-02 |
|         | Rank      | 1.000            | 3.000     | 2.000     | 5.000     | 4.000     | 9.000     | 6.000     | 8.000     | 7.000     | 10.000    |
| FIR8    | Mean      | <b>3.602e-03</b> | 9.224e-03 | 2.379e-02 | 7.485e-03 | 2.955e-02 | 9.750e-02 | 3.098e-02 | 8.786e-02 | 1.475e-01 | 2.478e-01 |
|         | Std       | 1.131e-03        | 3.131e-03 | 2.381e-02 | 2.647e-03 | 1.350e-02 | 5.493e-02 | 1.170e-02 | 3.935e-02 | 1.006e-01 | 7.542e-02 |
|         | Rank      | 1.000            | 3.000     | 4.000     | 2.000     | 5.000     | 8.000     | 6.000     | 7.000     | 9.000     | 10.000    |
| AvgRank | Rank      | <b>1.000</b>     | 3.875     | 4.500     | 3.125     | 2.625     | 8.625     | 6.750     | 8.625     | 6.375     | 9.500     |

the lowest mean fitness in all eight cases against every competitor. These two resolutions agree on the direction of the effect, while the block-level post hoc test determines whether the rank separation is sufficiently large relative to the complete method set.

The global block analysis in Table 16 shows an exceptionally strong omnibus separation ( $W = 0.917$ ). The Friedman-Holm post hoc comparisons are significant against RLTLBO, QQLMPA, GWO, FDB-TLABC, and FTO. They are not significant against MPHB, HBA, MPA, or EODE after correction across all competitors, despite A-C1 winning all eight cases and the instance-level Wilcoxon tests favoring A-C1 in every case against the internal references and in six of eight cases against EODE. This difference reflects the limited number of independent FIR cases and the conservativeness of the multiple-comparison procedure.

Together, the external FIR results provide the strongest directional evidence in favor of MPHBS within the evaluated FIR cases. A-C1 obtains the best mean fitness in all eight cases, the best average rank, and a very large omnibus concordance effect. The post hoc evidence is decisive against the lower-ranked external methods but more conservative against EODE and the internal references because only eight independent cases are available. Therefore, the FIR comparison supports a strong structured-domain advantage while avoiding a claim of universal pairwise significance.

## 6. Discussion

The evidence suggests that MPHBS should not be interpreted as a universally dominant optimizer, but as a structurally effective SARSA-guided information-exchange architecture whose benefit depends on the interaction between the search landscape and the way candidate memories are recombined. The empirical validation combines sensitivity analysis, ablation studies, convergence behavior, and external comparisons, so the interpretation separates configuration selection from final external assessment. This interpretation is important because the proposed contribution is not merely that HBA and MPA are executed together. The value of MPHBS lies in regulating how information produced by both populations is ranked, selected, recombined, and accepted during the search.

From an architectural perspective, MPHBS contributes a learning-guided ensemble between HBA and MPA through a mediator that operates on ranked candidate memories. The tensorized decision representation connects each action with a problem dimension, a ranked candidate index, and a source population, allowing source-aware component selection rather than global population mixing. The proxy-screened sub-exploration stage then performs component-wise hybrid-vector assembly at low cost, while only one selected candidate is

**Table 15**

Case-wise W/T/L results for the FIR external comparison. Each cell reports the wins/ties/losses of the column method against the final FIR configuration of MPHBS, A-C1, over 30 paired runs.

| Case | MPHB   | HBA    | MPA    | EODE    |
|------|--------|--------|--------|---------|
| FIR1 | 2/0/28 | 4/0/26 | 3/0/27 | 12/0/18 |
| FIR2 | 1/0/29 | 3/0/27 | 2/0/28 | 12/0/18 |
| FIR3 | 0/0/30 | 1/0/29 | 2/0/28 | 9/0/21  |
| FIR4 | 0/0/30 | 1/0/29 | 4/0/26 | 3/0/27  |
| FIR5 | 0/0/30 | 3/0/27 | 1/0/29 | 4/0/26  |
| FIR6 | 0/0/30 | 4/0/26 | 8/0/22 | 8/0/22  |
| FIR7 | 0/0/30 | 5/0/25 | 0/0/30 | 1/0/29  |
| FIR8 | 1/0/29 | 1/0/29 | 1/0/29 | 0/0/30  |

| Case | RLTLBO | QQLMPA | GWO    | FDB-TLABC | FTO    |
|------|--------|--------|--------|-----------|--------|
| FIR1 | 0/0/30 | 0/0/30 | 0/0/30 | 1/0/29    | 0/0/30 |
| FIR2 | 0/0/30 | 1/0/29 | 0/0/30 | 1/0/29    | 0/0/30 |
| FIR3 | 0/0/30 | 1/0/29 | 0/0/30 | 0/0/30    | 0/0/30 |
| FIR4 | 0/0/30 | 0/0/30 | 0/0/30 | 1/0/29    | 0/0/30 |
| FIR5 | 0/0/30 | 0/0/30 | 0/0/30 | 0/0/30    | 0/0/30 |
| FIR6 | 0/0/30 | 0/0/30 | 0/0/30 | 1/0/29    | 0/0/30 |
| FIR7 | 0/0/30 | 0/0/30 | 0/0/30 | 0/0/30    | 0/0/30 |
| FIR8 | 0/0/30 | 0/0/30 | 0/0/30 | 0/0/30    | 0/0/30 |

evaluated with the expensive objective function. Together with the dimension-wise guided mutation and component-replacement mechanism, this design allows useful partial information to be reused without forcing a full-solution inheritance from either parent optimizer. The conservative SARSA-style update further limits learning to the actions involved in the proxy-winning candidate and to the active dimensions, which keeps the mediator informative without replacing the underlying search dynamics of HBA and MPA.

This focused role of the learning component is central to the design. MPHBS does not distribute reinforcement learning throughout the full population update, nor does it use the proxy as a replacement for the objective function. Instead, the mediator is restricted to the information-exchange stage, where it regulates source-aware component reuse over ranked HBA and MPA memories. This keeps the learning mechanism interpretable while preserving the original search dynamics of both base optimizers.

The comparison with recent IPA and UAV path-planning methods further clarifies the methodological scope of MPHBS. Levy-flight-assisted IPA improves stochastic diversification, GINT-IPA improves the initial search state and IPA treatment process through problem knowledge, tBaF exploits path geometry with multi-threaded search bounds, MULIPA and DUALIPA exchange information across populations, and IPA-based UCAV optimization demonstrates the value of metaheuristics in engineering path planning. MPHBS instead focuses on learning-guided information exchange inside a domain-agnostic continuous optimization architecture. In particular, tBaF directly exploits the geometry of the UAV path-planning problem, whereas MPHBS does not assume an explicit geometric model and learns how to reuse ranked components produced by HBA and MPA. These directions are complementary rather than mutually exclusive: problem-informed initialization or geometry-aware candidate generation can provide strong initial structures, while a SARSA-style mediator can regulate how useful partial components from such structures are reused during subsequent continuous optimization.

The sensitivity and ablation results show that this mechanism is domain-dependent. CEC2022 selects B-C1 as the final CEC2022 configuration of MPHBS, whereas FIR selects A-C1 as the final FIR configuration of the same architecture. This difference should not be read as evidence for two different methods. Instead, both configurations instantiate MPHBS under domain-specific validated settings. In CEC2022, placing the FAD mechanism before the mediator appears to favor diversity before information exchange, which is useful in heterogeneous continuous landscapes. In FIR, placing FAD after the mediator appears

more beneficial because the algorithm first assembles structured candidate information and then perturbs the resulting components. The mechanism is therefore common, while the most effective placement and hyperparameter setting depend on the structure of the optimization landscape.

The CEC2022 external comparison supports a sober interpretation. B-C1 obtains the best mean-based average rank, and the significant Friedman test with  $W = 0.626$  confirms substantial global separation among methods. However, the post hoc analysis places B-C1 in the same leading statistical group as MPHB, MPA, EODE, QQLMPA, and FDB-TLABC, while identifying significant advantages over HBA, RLTLBO, GWO, and FTO. Thus, the CEC2022 evidence is best described as competitive robustness rather than universal dominance.

The FIR results provide stronger directional evidence in favor of MPHBS. A-C1 is the final FIR configuration of the same architecture, and it obtains the best mean fitness in all eight filter-design cases, the best average rank, favorable W/T/L behavior, and a very large omnibus concordance effect ( $W = 0.917$ ). The corrected post hoc analysis is decisive against the lower-ranked external methods but more conservative against EODE and the internal references because the benchmark contains only eight independent cases. The advantage of A-C1 nevertheless suggests that the complete information-exchange architecture is especially useful when informative fragments survive in the ranked candidate memories and can be recombined without destroying the spectral structure of the solution.

The comparison with RANDOM configurations and MPHB also clarifies the role of SARSA. The mediator should not be presented as a universal or automatic improvement over random exchange in every domain. In FIR, the advantage over RANDOM is more moderate in the ablation stage, whereas the advantage over MPHB, HBA, and MPA is much clearer. This indicates that the hybrid backbone is already strong and that the main contribution of MPHBS is the intelligent regulation of information exchange on top of that backbone. In other words, SARSA does not replace the search ability of HBA and MPA; it guides when and how ranked components from both populations should be reused. Although FIR provides the strongest empirical evidence, the relevance of the proposed mechanism is broader: it offers a general way to regulate information exchange between heterogeneous continuous optimizers when useful partial components persist in candidate memories. The results should therefore not be interpreted as evidence of universal dominance, but as evidence that the proposed information-exchange architecture, coordinated by a SARSA-style mediator, can provide competitive robustness in general-purpose landscapes and stronger gains in structured domains where useful partial components can be recombined.

## 7. Threats to validity

Several aspects should be considered when interpreting the results. First, the final settings are domain-specific. B-C1 was selected for CEC2022 and A-C1 was selected for FIR through internal validation, so direct transfer of either configuration to a new domain should not be assumed without additional validation. This does not weaken the MPHBS architecture itself; rather, it shows that the architecture is sensitive to benchmark structure and that automatic configuration transfer and adaptive FAD placement remain open issues. Because the selected configurations were obtained through internal validation, their generalization should also be evaluated on additional domains.

Second, the CEC2022 experiments were conducted at dimension  $D = 20$ . This setting supports controlled comparison under the adopted CEC2022 protocol, but it does not fully characterize scalability to higher-dimensional search spaces. Additional dimensions should therefore be considered in future work.

Third, the effectiveness of SARSA-guided recombination may depend on whether useful partial components survive in the ranked

**Table 16**

FIR external comparison using cases as independent blocks. W/T/L is reported from the anchor perspective using case-wise means. Friedman  $p = 9.103 \times 10^{-11}$  and Kendall's  $W = 0.917$ . Post hoc  $p$ -values are Holm-adjusted across competitors. Sig. inst. A/C reports the numbers of case-level paired Wilcoxon tests that remain significant after domain-wide Holm correction in favor of the anchor or competitor. Lower AvgRank is better.

| Method    | Role     | W/T/L | Anchor rank | Comp. rank | Holm $p$  | Reject | Sig. inst. A/C |
|-----------|----------|-------|-------------|------------|-----------|--------|----------------|
| MPHB      | BASELINE | 8/0/0 | 1.000       | 3.875      | 1.726e-01 | 0      | 8/0            |
| HBA       | PARENT   | 8/0/0 | 1.000       | 4.500      | 8.311e-02 | 0      | 8/0            |
| MPA       | PARENT   | 8/0/0 | 1.000       | 3.125      | 3.208e-01 | 0      | 8/0            |
| EODE      | EXTERNAL | 8/0/0 | 1.000       | 2.625      | 3.208e-01 | 0      | 6/0            |
| RLTLBO    | EXTERNAL | 8/0/0 | 1.000       | 8.625      | 3.785e-06 | 1      | 8/0            |
| QQLMPA    | EXTERNAL | 8/0/0 | 1.000       | 6.750      | 8.741e-04 | 1      | 8/0            |
| GWO       | EXTERNAL | 8/0/0 | 1.000       | 8.625      | 3.785e-06 | 1      | 8/0            |
| FDB-TLABC | EXTERNAL | 8/0/0 | 1.000       | 6.375      | 1.922e-03 | 1      | 8/0            |
| FTO       | EXTERNAL | 8/0/0 | 1.000       | 9.500      | 1.770e-07 | 1      | 8/0            |

candidate memories. The mechanism is designed to exploit dimension-wise information distributed across HBA and MPA populations, but this assumption may be less effective in landscapes where meaningful partial structures are rapidly destroyed, highly epistatic, or poorly reflected by rank-based memories. This is consistent with the heuristic component-wise justification of the mediator: the argument is strongest when useful partial contributions can be recombined without being dominated by destructive cross-dimensional interactions. A related limitation is that the proxy terms are not globally normalized and may not fully capture high-order interactions among dimensions. However, MPHBS does not use the proxy as a replacement for the objective function. The proxy only selects a candidate for evaluation within each episode, whereas the SARSA reward, acceptance decision, and global-best update are computed from the true fitness. This separation reduces the risk that proxy-scale bias directly affects the final optimization state. This threat is particularly relevant when moving from structured problems such as FIR design to less regular black-box domains.

Fourth, the external comparison depends on the selected baselines and their parameterization. EODE is included as a strong recent hybrid DE-based external comparator, RLTLBO and QQLMPA provide additional learning-based metaheuristic references, and GWO, FDB-TLABC, and FTO provide further reference points with different search mechanisms. Nevertheless, other DE-family methods or more heavily tuned configurations could be explored in future work. The present comparisons should therefore be interpreted under the adopted reproducible protocol rather than as an exhaustive ranking of all possible metaheuristic implementations.

Fifth, the FIR benchmark is a custom structured signal-processing benchmark. The eight cases, filter order, frequency specifications, fast Fourier transform (FFT) resolution, and weights define a controlled evaluation, but other FIR settings or objective formulations may produce different relative behavior. For this reason, the full FIR definition, source code, scripts, and generated artifacts are included in the public Zenodo reproducibility package associated with this study [74].

Sixth, the global nonparametric analysis is based on 12 CEC2022 functions and eight FIR cases as independent blocks. This avoids pseudoreplication from pooling repeated runs across heterogeneous problems, but it also limits the power of post hoc comparisons, particularly in FIR. Accordingly, non-rejection against a close competitor should not be interpreted as evidence of equivalence; the conclusions are based jointly on average ranks, Kendall's  $W$ , instance-level W/T/L patterns, corrected within-instance tests, and the Friedman–Holm results.

Finally, computational cost should be interpreted under matched function-evaluation budgets and not only through wall-clock time. The runners use equal function-evaluation budgets, population sizes, run counts, and common seed schedules, which is appropriate for paired black-box optimization comparisons. However, implementation details, vectorization, parallel execution, and objective-function overhead can affect measured runtime. The most reliable comparison in this study is therefore the function-evaluation-controlled solution quality, supported by paired statistical analysis and convergence curves.

## 8. Reproducibility statement

All experiments were implemented in MATLAB 2024b and executed on a Mac mini with an Apple M4 Pro processor, 24 GB of RAM, and macOS 26.3.1. The experimental protocol used 30 runs per method and problem instance under a common seed schedule that preserves run-level pairing across methods. For CEC2022, the experiments used the official `cec22_test_func` interface, dimension  $D = 20$ , population size  $N = 30$ , search bounds  $[-100, 100]$ , 12 benchmark functions, and a maximum budget of 10,000 function evaluations. Direct objective-call accounting produced 199,614–200,000 FEs across the final CEC2022 campaigns. Reported CEC2022 values correspond to residual errors with respect to the known optima.

For the FIR benchmark, the filter order was fixed to 30, resulting in  $D = 31$  decision variables bounded in  $[-1, 1]$ . The benchmark contains eight cases covering low-pass, high-pass, band-pass, and band-stop designs. The frequency response was evaluated using  $N_{fft} = 2048$ , and all compared methods were executed with population size  $N = 30$ , 30 paired runs, and a maximum budget of 150,000 function evaluations. Direct objective-call accounting produced 149,718–150,000 FEs across the final FIR campaigns. The objective function was defined as  $F(h) = W_p \text{MSE}_p + W_s \text{MSE}_s$ , with passband and stopband sets explicitly determined by each case definition.

The final MPHBS configurations were not selected during the external comparison stage. Instead, B-C1 was fixed as the final CEC2022 configuration of MPHBS and A-C1 was fixed as the final FIR configuration of MPHBS based on the preceding internal sensitivity and ablation analyses. External comparisons were then performed against MPHB, HBA, MPA, EODE, RLTLBO, QQLMPA, GWO, FDB-TLABC, and FTO under matched run counts, common seed schedules, and evaluation budgets. The statistical analysis used instance-level paired Wilcoxon signed-rank tests, domain-wide Holm correction, Friedman tests with functions or cases as blocks, Kendall's  $W$ , anchor-based Friedman–Holm post hoc comparisons, average ranks, and W/T/L summaries.

The experimental runners were designed to export raw run-level data, summary CSV files, LaTeX tables, convergence curves, checkpoints, and final workspaces. To facilitate independent verification, the MATLAB implementation, benchmark definitions, raw run-level results, generated CSV files, LaTeX tables, convergence figures, checkpoints, and workspaces are publicly available in the Zenodo reproducibility package associated with this study [74]. These artifacts allow the reported tables and figures to be regenerated from the same experimental outputs and independently verified.

## 9. Conclusion

This paper introduced MPHBS, a hybrid HBA–MPA metaheuristic whose central component is an information-exchange architecture coordinated by a SARSA-style mediator between the two base search processes. The proposed method uses tensorized ranked memories and

proxy-screened component-wise hybrid-vector assembly to exploit useful partial information distributed across both populations while evaluating only one expensive candidate per mediator episode. In this sense, MPHBS is not simply a parallel combination of HBA and MPA, but an architecture for regulating how candidate information is reused through ranked memory, source-aware selection, and dimension-wise replacement.

The internal validation selected B-C1 as the final CEC2022 configuration of MPHBS and A-C1 as the final FIR configuration of MPHBS. These configurations are domain-specific settings of one algorithmic architecture. On CEC2022, B-C1 obtains the best mean-based average rank and belongs to the leading statistical group, with significant post hoc advantages over HBA, RLTLBO, GWO, and FTO. On FIR, A-C1 obtains the best mean fitness in all eight cases and a very large global concordance effect, while corrected post hoc significance is concentrated against the lower-ranked external methods. The evidence therefore supports competitive robustness on CEC2022 and a strong directional advantage within the structured FIR benchmark without implying universal pairwise dominance.

Future work should study automatic configuration transfer, adaptive selection of the FAD placement, additional benchmark and real-world domains, deeper analysis of landscape-dependent behavior, and the integration of SARSA-style component selection with problem-informed mechanisms such as greedy initialization, geometry-aware search, hill-climbing local refinement, Deep Q-learning for larger action spaces, or population-level information transfer. These directions are important for understanding when ranked component memories preserve useful partial information and how SARSA-guided recombination can be adapted without requiring separate manual validation for each new domain.

#### CRedit authorship contribution statement

**Emilio Flores:** Writing – review & editing, Writing – original draft, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rodrigo Olivares:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Conceptualization.

#### Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the authors used OpenAI ChatGPT and Codex to assist with language editing, LaTeX formatting, manuscript organization, and reproducibility checks. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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#### Data availability

Reproducible data and code are available on Zenodo: <https://doi.org/10.5281/zenodo.21326361>.

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